

Extreme Temperatures and Adaptive Health Investment: Evidence from Sanitation Behaviors in India

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Abstract

Extreme temperatures are known to negatively affect health in the short run, yet their persistent effects remain underexplored. This paper examines whether extreme temperatures encourage adaptive investments in health technologies over time. Exploiting year-to-year variation in temperature within districts in rural India, I find that cold and hot days persistently increase the latrine stock over multiple years. Heterogeneity analyses suggest a discomfort channel, whereby households construct latrines to avoid walking outside for open defecation under extreme temperatures. The health benefits from this increased latrine investment are comparable in magnitude to existing estimates of the negative health impacts of extreme temperatures.

JEL: I15, O13, Q54, Q56

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1 Introduction

Policymakers and researchers increasingly recognize the substantial negative impacts of climate change on human welfare. Climate change increases the frequency of extreme weather events, which in turn reduce human welfare by increasing mortality (e.g., Deschênes and Greenstone, 2011) and reducing agricultural productivity (e.g., Schlenker and Roberts, 2009) and labor productivity (e.g., Somanathan et al., 2021). These short-run negative welfare consequences are well documented.

However, little is known about whether weather shocks can have persistent positive effects on human welfare. I document that temperature shocks can induce adaptive investments in indoor health technologies by discouraging outdoor health behaviors that entail greater discomfort under extreme temperatures. When these outdoor behaviors pose health risks, weather-induced investments in indoor health technologies can ultimately improve health over time through sustained use. This focus on persistent positive effects, driven by adaptive health investments, offers a new perspective that differs from the growing literature emphasizing persistent negative effects.¹

This paper examines the persistent effect of extreme temperatures on health investment by investigating sanitation behaviors, specifically the construction of latrines, which are durable goods used over time. Extreme temperatures can affect a household’s decision of whether to construct latrines or to continue open defecation in two opposing ways.

First, extreme temperatures can have a positive effect on latrine investment by increasing the discomfort associated with open defecation (discomfort channel). Open defecation requires people to walk outside to a place far from home, and extreme temperatures can intensify the discomfort of this outdoor behavior. The heightened discomfort under extreme temperatures may discourage people from open defecation, thereby motivating them to invest in latrines as a means of adaptation.

Conversely, extreme temperatures, particularly extremely high temperatures, can have a negative effect on latrine investment by reducing income (income channel). Extreme heat has been shown to negatively affect income by reducing agricultural productivity (Burgess et al., 2017; Colmer, 2021). This reduction in income can tighten financial constraints, making it more difficult for households to afford latrine construction, thus discouraging this form of health investment.

I empirically examine which of these two channels, the discomfort channel or the income channel, dominates. If the positive effect through the discomfort channel dominates, latrines

¹ A small but growing number of studies have examined the persistent effects of weather shocks, primarily highlighting their negative impacts on economic growth (Dell et al., 2012; Foreman, 2020) and educational attainment (Park, 2022).

constructed in response to extreme temperatures may remain in use for many years because they are durable goods, generating a persistent effect on sanitation investment.

I examine temperature-induced latrine investment in the context of India’s nationwide sanitation policy, the Swachh Bharat Mission (SBM), which started in 2014. Under this policy, the Indian government provided subsidies of up to INR 12,000 per household for latrine construction in rural areas, covering most of the initial cost of a basic latrine. In this study context, the discomfort channel is expected to be more influential than the income channel, which is attenuated by the subsidy. My empirical analysis combines administrative data on the district-level number of latrines under the SBM with raster data on daily temperature and rainfall, both covering the period from 2012 to 2019.

To examine the causal effect of temperature on latrine investment, I exploit presumably random year-to-year variation in temperature at the district level after controlling for district fixed effects, state-by-year fixed effects, and rainfall. I group the daily temperature measures into eight bins to investigate the nonlinear relationship between temperature and latrine investment. I also employ a distributed lag model that includes up to three years of lagged temperature and test the persistence of the effect by examining whether the lagged effects are positive. I then examine whether the cumulative effect, defined as the sum of contemporaneous and lagged effects, is positive or negative, indicating which channel dominates.

I find that extreme temperatures increase latrine investment, and this positive effect persists over multiple years. Specifically, a sustained increase of one additional cold day per year with an average temperature below 5°C or 5–10°C raises the stock of latrines after three years by 21.1 or 17.1 per 1,000 households, respectively, relative to a day in the 15–20°C reference range. This cumulative effect amounts to a 7.9% or 6.4% increase, respectively, relative to the pre-SBM mean of the latrine stock. Conversely, I find smaller positive effects of higher temperatures, as the negative effect of the income channel can offset the positive effect of the discomfort channel. For example, one additional hot day per year with an average temperature of 25–30°C or 20–25°C raises the stock by 4.6 or 6.4 per 1,000 households (a 1.7% or 2.4% increase), respectively. The effects become imprecise for temperatures exceeding 30°C, as these temperature ranges are more susceptible to the income channel. Overall, the positive cumulative effects, observed across most temperature bins, suggest that the discomfort channel dominates the income channel. The persistence of this effect is evident particularly at lower temperatures, where the lagged estimates remain positive and mostly statistically significant one to three years after the shock, indicating that the stock of latrines stays elevated rather than reverting to its previous level.

A variety of robustness checks corroborate my findings on the positive and persistent

effects of extreme temperatures on latrine investment. Specifically, the results are robust to multiple checks: varying the number of lagged years from 1 to 10 in the distributed lag model, addressing potential measurement errors in the outcome, accounting for baseline latrine coverage that may affect subsequent latrine construction, and estimating an alternative long-difference specification that exploits medium-run temperature variation.

Further heterogeneity and temperature deviation analyses provide supporting evidence for the discomfort channel. A heterogeneity analysis by baseline temperatures shows that the positive effects of high temperatures on latrine investment are pronounced in districts with lower baseline temperatures, but much weaker in those with higher ones. These differential effects are consistent with the discomfort channel: people in colder climates are less adapted to high temperatures and therefore invest in latrines to mitigate the greater discomfort caused by heat. An alternative specification using temperature deviations from historical means (standardized temperature shocks that are comparable across districts) also demonstrates positive effects on latrine investment, consistent with the baseline findings.

Moreover, heterogeneity analyses by crop area and by SBM policy period suggest the presence of an income channel that offsets the positive effect of the discomfort channel. A heterogeneity analysis by baseline crop area shows that the positive effects of higher temperatures are less pronounced in districts with larger crop areas, suggesting that the income channel, operating through reduced agricultural output, exerts a negative effect at high temperatures. A complementary heterogeneity analysis by SBM policy period further supports the income channel: the positive effects of extreme temperatures on latrine investment emerge strongly in the post-SBM period, when subsidies eased households' financial burden, but are largely absent in the pre-SBM period, when households faced substantially higher out-of-pocket costs.

Conversely, I find that extreme temperatures generally do not affect latrine use at the intensive margin following construction, with the exception of very high temperatures. To examine the effect of temperature on latrine use conditional on ownership, I use a household-level panel dataset from two survey rounds conducted in 2013–2014 and 2018, covering 120 villages across four northern Indian states where open defecation is widely prevalent (Coffey et al., 2014; Gupta et al., 2019). Across most temperature ranges, temperature does not influence latrine use, whether temperature exposure is measured over periods as short as a week or as long as a year. This is likely due to the high baseline latrine use rate, which averages 79%, as well as the infrequent occurrence of cold days and the population's adaptation to high temperatures in these predominantly hot states. However, I find that extremely hot days (above 35°C) increase latrine use in the short run, suggesting the discomfort channel also plays a role in driving latrine use at the intensive margin.

Taken together, my analysis highlights that extreme temperatures can encourage adaptive investment in health technologies by increasing the discomfort of outdoor behaviors. Because latrine construction can reduce diarrheal child mortality (Motohashi, 2026), temperature-induced latrine investment may generate long-lasting health benefits. A back-of-the-envelope calculation suggests that a sustained increase of one additional cold or hot day per year could reduce the diarrheal post-neonatal mortality rate in rural India by 0.13–0.73% through the resulting increase in the latrine stock. These health benefits are comparable in magnitude to estimates of the negative health impacts of extreme temperatures in the previous literature, including those reported by Burgess et al. (2017).

This paper makes three contributions. First, I contribute to the literature on the consequences of climate change by showing that weather shocks can induce persistent adaptive investments in health technologies, with a potential co-benefit for health. Most past studies demonstrate the short-run effects (level effects) of weather shocks on human health (Deschênes and Greenstone, 2011; Burgess et al., 2017; Lepault et al., 2026), educational outcomes (Garg et al., 2020), agricultural productivity (Schlenker and Roberts, 2009; Colmer, 2021), and labor productivity (Adhvaryu et al., 2020; Somanathan et al., 2021; Heyes and Saberian, 2022), which dissipate once these shocks subside. However, a growing literature shows that weather shocks can have persistent negative effects (growth effects) on economic growth (Dell et al., 2012; Foreman, 2020) and human capital accumulation (Park, 2022). I complement these studies by documenting a mechanism through which weather shocks can have persistent positive effects: shifting households away from harmful outdoor behaviors and toward health-improving durable goods used indoors. Moreover, unlike much of the adaptation literature, which examines how adaptation, as captured by air conditioning, income, or the local climate, attenuates the direct effects of extreme temperatures on health (Barreca et al., 2016; Heutel et al., 2021; Carleton et al., 2022), I treat the adaptive response itself as the outcome and provide evidence that latrines constructed to avoid the discomfort of open defecation under extreme temperatures may generate an unintended co-benefit for health.

Second, I contribute to the literature on technology adoption in developing countries by showing that weather can be another major determinant of technology adoption. In these countries, outdoor health behaviors, including open defecation (Cameron et al., 2022), the collection of unsafe water (Kremer et al., 2011), and the collection and use of biomass for cooking (Hanna et al., 2016), are widespread. These outdoor behaviors are closely linked to water and air pollution, leading to substantial health risks for households with limited coping measures. Past studies have shown that subsidies, credit, and information campaigns can encourage the adoption of health-improving technologies such as latrines (Guiteras et al.,

2015), piped water (Devoto et al., 2012), and clean cooking fuels (Imelda, 2020), thereby reducing the dependence on these harmful outdoor behaviors. I complement these studies by showing that weather shocks also drive health technology adoption, which can in turn discourage outdoor behaviors detrimental to human health.

Lastly, I contribute to the behavioral economics literature on projection bias in forward-looking decisions by providing evidence from a developing-country context that is consistent with such bias. Past studies in developed countries show that current weather conditions can disproportionately influence decisions with future consequences, including purchases of cold-weather items and cars, as well as college enrollment decisions (Conlin et al., 2007; Simonsohn, 2010; Busse et al., 2015). In the same vein, I show that short-run year-to-year temperature shocks affect latrine construction, even though latrines are durable goods used for many years. Although rational households would decide whether to construct latrines based on expected future climate conditions and the associated long-run discomfort of open defecation, my findings suggest that these decisions are instead influenced by short-run temperature shocks. This evidence from a developing-country context is important, as such bias may be particularly relevant where access to information about future climate and weather conditions is limited.

The remainder of this paper is organized as follows. Section 2 discusses the relationship between temperature and sanitation behaviors, as well as the study setting in rural India. Sections 3 and 4 describe my data and empirical strategy, respectively. Section 5 presents the effects of temperature on latrine investment, and Section 6 investigates the underlying mechanisms. Section 7 examines the intensive-margin effect on latrine use. Finally, Section 8 concludes the paper.

2 Temperature and Sanitation Behaviors in India

I present two main channels through which extreme temperatures can affect sanitation behaviors. I then discuss how these channels operate in the context of this paper: the Swachh Bharat Mission, a nationwide sanitation policy in rural India.

2.1 Temperature and Sanitation Behaviors

Extreme temperatures can have two opposing effects on latrine investment and use: (i) a positive effect through the discomfort channel and (ii) a negative effect through the income channel.²

² Alternative channels related to construction feasibility and government relief are discussed in Section 6.3. This section focuses on the two primary channels examined in this paper.

First, extreme temperatures can increase latrine investment and use by intensifying the discomfort associated with open defecation (discomfort channel). Open defecation requires people to walk outside to a place far from home, and extreme temperatures can make this outdoor practice more uncomfortable. The heightened discomfort may discourage people from practicing open defecation, thereby motivating them to construct and use latrines as a means of adaptation.³ This discomfort channel is consistent with public health studies that document seasonal variation in latrine use (Routray et al., 2015; Sinha et al., 2017). These studies find that latrine use is more likely during the dry cold season and the rainy season, suggesting that people avoid walking outside for open defecation when the weather is uncomfortable. Moreover, the discomfort channel is evident in the reasons that households give for constructing latrines in the Rural Sanitation Surveys, the household survey data used in Section 7 (Coffey et al., 2014). When asked in an open-ended question why they constructed a latrine, 52% of respondents in latrine-owning households mention protection from weather, the second most frequently mentioned specific reason after saving time and ahead of any reason related to women or health, which is consistent with the discomfort of practicing open defecation outdoors in harsh weather. The same pattern holds among households that have not yet constructed a latrine: when asked about the benefits of having a latrine, 49% of respondents who practice open defecation in households without a latrine mention protection from weather, again second only to saving time.

Conversely, extreme temperatures, particularly extremely high temperatures, can decrease latrine investment and use by reducing income (income channel). Extreme heat has been shown to reduce income by lowering agricultural productivity (Burgess et al., 2017; Colmer, 2021). This reduction in income can tighten financial constraints, making it more difficult for households to afford latrine construction and thus discouraging this form of health investment. Reduced income can also depress latrine use by making it harder to cover maintenance costs, including the costs of hiring desludging tankers and laborers to regularly empty pits and septic tanks. These two opposing channels are formally presented in a simple conceptual framework in Appendix A.

My empirical analyses in Sections 4 and 5 capture the net effect of the discomfort and income channels. The sign of this net effect indicates which channel dominates: a positive (negative) net effect suggests that the discomfort (income) channel outweighs the other. Additionally, I conduct heterogeneity analyses in Section 6 to examine each channel.

³ Evidence from the United States is also consistent with this discomfort channel, whereby extreme temperatures discourage outdoor activities: attendance at outdoor baseball games declines on both very hot and very cold days, whereas attendance at covered stadiums is unaffected (Kuruc et al., 2025).

2.2 Study Setting: The Swachh Bharat Mission in India

During my study period, the Indian government subsidized latrine construction across rural India under the SBM, aiming to eliminate open defecation. In this setting, the discomfort channel is expected to dominate the income channel, which the subsidy attenuates.

Historically, many people in India have practiced open defecation, which harms child health by increasing diarrheal disease and child mortality. To address this issue, the Indian government launched the SBM nationwide in 2014, offering subsidies for latrine construction that are disbursed after the constructed latrine is verified. The subsidy is capped at INR 12,000 (approximately USD 125 at 2026 exchange rates) per household, covering most of the initial cost of a basic latrine in rural India. This policy resulted in the construction of over 100 million household latrines across the country by the policy deadline in 2019 (Motohashi, 2026).⁴

This SBM context suggests that the negative effect of extreme temperatures via the income channel is limited during my study period, because the subsidy relaxes households' financial constraints on latrine construction.⁵ Therefore, I expect that the discomfort channel dominates the income channel, leading to a net positive effect of extreme temperatures on latrine investment. Within my sample period, this prediction should hold more strongly in the post-SBM years (2014–2019) than in the pre-SBM years (2012–2013), when households faced higher out-of-pocket costs and the income channel was likely more binding.

The SBM context also shapes the interpretation of my estimates. While the SBM has increased latrine construction throughout rural India, the magnitude of this increase may vary with districts' exposure to extreme temperatures. My analysis therefore examines whether districts experiencing more extreme temperatures see a larger increase in the number of latrines.

3 Data

To examine the effect of temperature on latrine investment, I use administrative data on the district-level number of latrines under the SBM, combined with raster data on daily temperature and rainfall from 2012 to 2019. I also use a household-level panel survey on rural sanitation in four northern Indian states to examine the effect of temperature on latrine

⁴ This paper studies household-level latrine construction in rural India, primarily subsidized under the SBM's rural component, rather than public toilets, which are more common in urban areas. This scope matches the discomfort and income channels, both describing a household's own decision to build a latrine.

⁵ The SBM subsidy is provided to households after verification of latrine construction, so credit constraints may still be relevant. Moreover, because the subsidy applies to construction costs, the income channel may remain relevant for maintenance costs associated with latrine use.

use.

3.1 Latrine Investment

The first outcome variable is latrine investment, measured as the cumulative number of constructed latrines. I use administrative data on the district-level number of household latrines in rural India recorded under the SBM from 2012 to 2019, as compiled in Motohashi (2026). Based on this dataset, I compute the cumulative number of household latrines per 1,000 households in each district and year, using the number of households in each district from the 2011 Census as the denominator.⁶

One concern about this dataset is that the number of latrines might be systematically over-reported, leading to a measurement error. This dataset is compiled by the Government of India under the SBM, which aimed to achieve 100% latrine coverage by 2019. Thus, over-reporting is more likely in years closer to the 2019 deadline. Hossain et al. (2022) validate this latrine dataset by comparing it with statistics from the 2015–2016 National Family Health Survey and find that it is reliable at least until 2016. Therefore, I conduct a robustness check in Section 5.2 that restricts the sample period to the years up to 2016.

3.2 Latrine Use

The second outcome variable is latrine use. I use household-level panel data of latrine use over two survey rounds (2013–2014 and 2018) from the Rural Sanitation Surveys (Coffey et al., 2014; Gupta et al., 2019). The Rural Sanitation Surveys track the same households over the two rounds in 157 villages across 11 districts in four northern Indian states (Rajasthan, Madhya Pradesh, Uttar Pradesh, and Bihar), where open defecation is widely prevalent.

From the Rural Sanitation Surveys dataset, I use each household’s latrine ownership status and each household member’s latrine use in each survey round.⁷ For the empirical analysis, I construct the household-level latrine use rate as the proportion of household members who use latrines.⁸

To match the Rural Sanitation Surveys dataset with the weather data, I also use village-level GPS coordinates. The surveyed villages’ locations are only available at an approximate

⁶ Because the denominator is fixed at the 2011 Census count while the numerator accumulates through 2019, the ratio can exceed 1,000 in districts where the households recorded by the SBM program outnumber those counted in the 2011 Census, which is the case in about 6% of districts by 2019. As shown in Appendix Figure C1, the results are robust to excluding these districts.

⁷ The Rural Sanitation Surveys ask about each member’s usual defecation practice (open defecation or latrine use).

⁸ The latrine use rate is calculated over household members who are older than two years and have lived in the house for more than two months in the past year; these members were asked about their latrine use in the survey.

0.25-degree resolution. Accordingly, when matching the Rural Sanitation Surveys dataset to the weather data, I use weather raster data within a 0.25-degree buffer around each village’s GPS coordinates. My analysis focuses on 120 villages, out of 157 villages, for which GPS information is available.

3.3 Weather

As the treatment variable, I use daily gridded temperature data at 1-degree resolution from the India Meteorological Department (IMD) (Srivastava et al., 2009). As a control variable, I also use daily gridded rainfall data at 0.25-degree resolution from the same IMD source (Pai et al., 2014).⁹ For the empirical analysis, I construct daily average temperature as the average of the maximum and minimum temperatures recorded in the IMD temperature dataset, expressed in degrees Celsius (°C).

To match these weather variables with the district-level dataset on latrine investment, I compute the district-level means of daily average temperature and rainfall based on the gridded datasets and the 2011 district boundaries. Similarly, for the Rural Sanitation Surveys dataset on latrine use, I compute the means of daily average temperature and rainfall inside the 0.25-degree buffer around each village’s GPS coordinates.

3.4 Data Matching and Sample Construction

For the analysis of the effect of temperature on latrine investment, I construct a balanced panel dataset on latrine construction and weather variables covering 614 districts from 2012 to 2019. I spatially match the district-level number of latrines and mean daily weather variables based on the 2011 district boundaries.¹⁰

For the analysis of the effect of temperature on latrine use, I construct a balanced panel dataset on latrine use and weather variables covering 1,188 households in 120 villages over the two survey rounds. I spatially match the household-level survey data with the village-level daily weather variables based on the village GPS coordinates.

Table 1 reports the summary statistics of all variables used in the analysis, and Figure 1 shows the distributions of daily average temperature.

⁹ These datasets are constructed by interpolating temperature measures from 395 stations and rainfall measures from 6,955 stations across India (Srivastava et al., 2009; Pai et al., 2014).

¹⁰ Following Motohashi (2026), I address changes in district boundaries by organizing all data according to the 2011 boundaries. Latrine data based on the 2019 boundaries are aggregated to the 2011 boundaries by accounting for the district splits between 2011 and 2019.

4 Empirical Strategy

I estimate the causal effect of temperature on latrine investment by exploiting year-to-year variation in temperature within districts. I test the persistence of this effect by examining the lagged effects in a distributed lag model that includes lagged temperatures, and I use the sign of the cumulative effect to examine which channel dominates.

Specifically, I adopt the following two-way fixed effects specification:

$$Latrine_{dt} = \sum_l \sum_j \beta_{jl}^{INV} BinTemp_{dtjl} + \sum_l \sum_k \delta_{kl}^{INV} DecileRain_{dtkl} + \eta_d + \nu_{st} + \varepsilon_{dt} \quad (1)$$

where $Latrine_{dt}$ is the cumulative number of latrines per 1,000 households in district d in year t . $BinTemp_{dtjl}$ is the number of days in which the average temperature ($^{\circ}\text{C}$) falls in the j th bin in district d in year $t - l$, where l denotes the lag in years and $l = 0$ corresponds to the contemporaneous year. $DecileRain_{dtkl}$ denotes the number of days in which rainfall falls in the k th decile in district d in year $t - l$.¹¹ This rainfall variable is included as a control to ensure that the estimated temperature effects are not confounded by rainfall. I include district fixed effects (η_d) to control for time-invariant unobserved district-level determinants of latrine construction (e.g., social norms regarding open defecation and long-run climate conditions), as well as state-by-year fixed effects (ν_{st}) to control for shocks unique to each state in each year (e.g., changes in state-level sanitation policies and economic conditions). Standard errors are clustered at the district level to address serial correlation and because temperature varies at the district level.

I define eight temperature bins in $BinTemp_{dtjl}$ as follows: $<5^{\circ}\text{C}$, $5\text{--}10^{\circ}\text{C}$, $10\text{--}15^{\circ}\text{C}$, $15\text{--}20^{\circ}\text{C}$, $20\text{--}25^{\circ}\text{C}$, $25\text{--}30^{\circ}\text{C}$, $30\text{--}35^{\circ}\text{C}$, and $>35^{\circ}\text{C}$. These 5°C bins are chosen to capture the nonlinear temperature-latrine relationship in a flexible way while retaining a sufficient number of observed days in each bin. The $15\text{--}20^{\circ}\text{C}$ bin, representing moderate temperatures, serves as a reference bin and is dropped from the regression. Thus, the coefficient for each temperature bin j (β_{jl}^{INV}) captures the effect of an additional cold or hot day in bin j on the number of latrines per 1,000 households, relative to a day in the $15\text{--}20^{\circ}\text{C}$ bin. I choose this reference bin because it has the smallest effect on latrine investment among all the bins in the baseline specification, consistent with discomfort being lowest at moderate temperatures.

This regression specification exploits presumably random year-to-year variation in temperature to estimate the causal effect of temperature on latrine investment. By including

¹¹ Because 57% of district-days record no rainfall, the cutoffs for the lowest five deciles all equal zero, and the ten decile bins collapse into six distinct bins: a single no-rainfall bin and the sixth through tenth decile bins. The no-rainfall bin serves as the reference category and is dropped from the regression.

district fixed effects (η_d) and state-by-year fixed effects (ν_{st}), the temperature effect is identified from district-specific deviations in temperature around district averages, after controlling for shocks common to all districts in a state. Because these year-to-year fluctuations in temperature are unpredictable and presumably random, the coefficients β_{jl}^{INV} can have a causal interpretation.

To estimate the persistence of the effect of temperature on latrine investment, I use a distributed lag model that includes lagged temperature. Specifically, the baseline specification includes temperature variables lagged by up to three years ($l \leq 3$), in addition to the contemporaneous term ($l = 0$). Because the outcome is the cumulative stock of latrines, each coefficient β_{jl}^{INV} captures the change in the stock of latrines l years after an additional day in bin j . I also compute the cumulative effect for each temperature bin by summing the estimated contemporaneous and lagged effects ($\sum_l \beta_{jl}^{INV}$), which measures the change in the stock of latrines three years after the start of a sustained increase of one additional day in bin j per year (that is, with the additional day in the current year and in each of the three preceding years). The three-year window is chosen because deciding on latrine construction, applying for the SBM subsidy, and completing the construction can take several years. However, the results are robust to changing the maximum number of lags from 1 to 10 years, as discussed in Section 5.2.

The coefficients of interest are β_{jl}^{INV} . The cumulative effect derived from β_{jl}^{INV} indicates which of the two channels dominates: if it is positive and statistically significant, this would suggest that the discomfort channel dominates the income channel. In turn, I assess persistence from the lagged coefficients themselves: if an additional day merely brought forward construction that households would have undertaken anyway, the stock would return to its counterfactual level within a few years and the coefficients on longer lags would fall to zero, whereas coefficients that stay positive through the end of the lag window indicate that the stock remains elevated.¹²

5 Results

5.1 Baseline Results

I find that both low and high temperatures increase latrine investment, and this positive effect persists over multiple years.

¹² Unlike in previous studies of flow outcomes such as economic growth, where the sum of the contemporaneous and lagged effects identifies persistent effects because temporary effects are offset by lagged effects of the opposite sign (Dell et al., 2012), the sum of the coefficients on a stock outcome adds up the stock differences at each horizon and therefore remains positive even if construction is merely brought forward.

In Figure 2, I find a U-shaped cumulative temperature-latrine relationship, with a steeper slope in the low-temperature bins. A sustained increase of one additional cold day per year with an average temperature below 5°C or 5–10°C raises the stock of latrines after three years by 21.1 or 17.1 per 1,000 households, respectively, relative to a day in the 15–20°C range (Column 1 of Table 2). This cumulative effect amounts to a 7.9% or 6.4% increase relative to the pre-SBM mean of the latrine stock.¹³ Conversely, I find smaller positive effects of higher temperatures on latrine investment. For example, one additional hot day per year with an average temperature of 20–25°C or 25–30°C raises the stock by 6.4 or 4.6 per 1,000 households (a 2.4% or 1.7% increase), respectively (Column 1 of Table 2).¹⁴

The larger effects of lower temperatures are consistent with the discomfort channel. People in rural India are generally less adapted to lower temperatures than to higher ones, as the country experiences a hot climate on average, with daily average temperatures highly concentrated in the 25–30°C range (Panel A of Figure 1). Consequently, the discomfort of walking outside for open defecation is greater at lower temperatures, which may drive a stronger motivation to invest in latrines.

Another factor contributing to the larger effects of lower temperatures could be the income channel. The negative effect of temperature on agricultural output has been shown to be concentrated at high temperatures in India (Burgess et al., 2017; Colmer, 2021). This negative effect of the income channel on latrine investment can offset the positive effect of the discomfort channel, thereby making the positive effects of higher temperatures smaller. The positive effects on latrine investment even become statistically insignificant at very high temperatures exceeding 30°C, where the negative income effects are expected to be more pronounced (Panel A of Figure 2 and Column 1 of Table 2). A more detailed examination of the income channel is presented in Section 6.2.

Moreover, the lagged coefficients show that the effect of temperature shocks on latrine investment persists over multiple years. As shown in Panel B of Figure 2, the estimates for the low-temperature bins remain positive and mostly statistically significant one to three years after the shock, indicating that the stock of latrines stays elevated rather than reverting to its previous level.¹⁵ The lagged estimates in the high-temperature bins are also mostly positive, and those for the 20–30°C bins remain statistically significant one to two years

¹³ To calculate the effect in percentage, I divide the estimated coefficient by the mean of the dependent variable in the pre-SBM period (2012–2013). I adopt the same approach for all the following results.

¹⁴ For reference, Appendix Table D1 presents the cumulative effects of the rainfall decile bins from the same specification, measured relative to a day without rainfall (the omitted reference category, which serves as a natural benchmark with no rain-related discomfort for outdoor activities). I find no systematic relationship between rainfall and latrine investment: the cumulative effects are statistically insignificant, except for the seventh and eighth decile bins, whose effects are negative but small relative to the temperature effects.

¹⁵ All estimates shown in Panel B of Figure 2 are also reported in Appendix Table D2.

after the shock, although the persistence is less pronounced than in the low-temperature bins. The persistence of these effects can be partly attributed to the durability of latrines: once constructed in response to a temperature shock, they remain part of the sanitation capital stock in subsequent years.

5.2 Robustness Checks

The results are robust to multiple checks: varying the number of lagged years in the distributed lag model, addressing potential measurement errors in the outcome, accounting for baseline latrine coverage that may affect subsequent latrine construction, and estimating an alternative long-difference specification that exploits medium-run temperature variation.

Number of Lagged Years.—While the baseline specification includes three years of lagged temperatures, I estimate the cumulative effect with the maximum number of lags ranging from 1 to 10 years to examine whether the results are sensitive to the choice of lag length. As shown in Appendix Figure C2 and Appendix Table D3, the estimated cumulative effects in the lower-temperature ranges remain positive and mostly statistically significant regardless of the number of lagged years.¹⁶ In the higher-temperature ranges, the cumulative effects for the 20–25°C bin remain positive and mostly statistically significant regardless of the number of lagged years, whereas those for the bins above 25°C are positive for up to three to four years of lags but become imprecise with longer lags, as the standard errors increase substantially with the number of lagged temperature variables. This pattern is consistent with the smaller baseline effects in these ranges, where the income channel can offset the discomfort channel.

Measurement Errors in the Outcome.—As discussed in Section 3.1, the number of latrines reported in the administrative dataset of the SBM is unlikely to suffer from measurement errors, at least until 2016. Therefore, I conduct a robustness check by estimating the regression using observations through 2016.

In Appendix Figure C3, I find that the cumulative effect remains positive and statistically significant for the coldest bins in the sample through 2016, although the estimates are smaller than those in the baseline specification for both lower and higher temperatures. The smaller estimates may reflect tighter financial constraints on latrine construction during this period. Subsidy disbursement under the SBM likely accelerated as the 2019 deadline for universal latrine coverage approached. Thus, through 2016, rural households in India likely faced more

¹⁶ These cumulative effects also grow with the number of lags: because the cumulative effect measures the stock change after as many years of a sustained increase of one additional day per year in a temperature bin as there are lags, each additional year adds to the stock as long as the lagged coefficients remain positive.

limited access to the subsidy scheme and bore a larger share of the construction cost, which would have attenuated their investment response to extreme temperatures, whether cold or hot. This explanation is consistent with the results of the heterogeneity analysis by SBM policy period in Section 6.2, which finds much weaker effects in the pre-SBM period.

Baseline Latrine Coverage Affecting Subsequent Latrine Construction.—During my study period, latrine coverage under the SBM increased substantially across rural India. Consequently, districts with lower baseline coverage may have experienced larger increases in latrine construction. If the occurrence of extreme temperatures is negatively correlated with baseline latrine coverage, my baseline results might merely be capturing the effect of this baseline coverage. To address this concern, I conduct a heterogeneity analysis comparing the effects of temperature in districts with pre-SBM baseline coverage (measured in 2013) above the sample median to those in districts with coverage below the median.

As shown in Appendix Figure C4, I find positive cumulative effects of extreme temperatures both in districts with higher baseline coverage and in those with lower baseline coverage. These similar results across the two groups suggest that my baseline estimates are not simply picking up the effect of baseline latrine coverage correlated with the occurrence of extreme temperatures.

Long-Difference Specification.—The baseline specification captures households’ adaptation to short-run weather shocks using year-to-year temperature variation. However, climate change unfolds over longer time horizons, and households’ adaptation to longer-run climate change may differ in nature from their adaptation to short-run weather variation. To examine this longer-run response, I adopt a complementary long-difference specification that exploits changes in temperature exposure between the early period (2012–2015) and the late period (2016–2019) of the sample. Given the limited sample period, however, the gap between the two periods remains relatively short.

Specifically, I estimate the following long-difference specification:

$$\Delta Latrine_d = \sum_j \beta_j^{LD} \Delta BinTemp_{dj} + \sum_k \delta_k^{LD} \Delta DecileRain_{dk} + \mu_s + \varepsilon_d \quad (2)$$

where each Δ variable denotes the change in the corresponding variable defined in regression 1, computed for each district as the difference between its mean in the late period (2016–2019) and its mean in the early period (2012–2015). μ_s represents state fixed effects, which I include in one specification and omit in the other. The 15–20°C bin serves as the reference bin, and robust standard errors are reported.

As shown in Appendix Figure C5 and Appendix Table D4, the long-difference estimates

preserve the U-shaped temperature-latrine relationship found in the baseline specification, with positive effects that are statistically significant across most temperature bins, regardless of whether state fixed effects are included. The effects of lower temperatures are broadly comparable in magnitude to the baseline cumulative effects, whereas the effects of higher temperatures are larger and more precisely estimated than in the baseline specification. These results suggest that the response to extreme temperatures remains present even when households' adaptation to medium-run climate change is more fully accounted for. If anything, the larger high-temperature estimates suggest that the discomfort-driven response may even strengthen when temperature exposure is measured over a longer horizon.

6 Mechanisms

The net positive effect of extreme temperatures on latrine investment documented in Section 5 suggests that the discomfort channel dominates the income channel. Further heterogeneity and temperature deviation analyses in this section provide supporting evidence for the discomfort channel, while also suggesting the presence of an offsetting income channel.

6.1 Mechanism 1: Discomfort Channel

To investigate the discomfort channel, I examine heterogeneous effects by baseline temperature levels and conduct an alternative analysis using temperature deviations from historical means. These analyses suggest that the positive effects of extreme temperatures are concentrated in districts less adapted to such conditions, and that the U-shaped temperature-latrine relationship holds when using standardized temperature shocks that are comparable across districts.

Heterogeneity Analysis by Baseline Temperatures.—I conduct a heterogeneity analysis, comparing effects in districts whose baseline temperatures, measured over the pre-study period (2002–2011), are above the sample median (25.7°C) with effects in districts whose baseline temperatures are below the median. The discomfort channel suggests that people experience greater discomfort from walking outside for open defecation when exposed to temperatures they are less adapted to. For example, people living in districts with a lower baseline temperature could be more sensitive to high-temperature shocks than people living in districts with a higher baseline temperature. Therefore, high-temperature shocks are expected to cause a larger increase in latrine investment in these cooler districts than in districts with a higher baseline temperature. Conversely, districts with a higher baseline temperature may respond more strongly to low-temperature shocks than these cooler districts do.

Consistent with the discomfort channel, I find that the positive effects of higher temperatures on latrine investment are concentrated in districts with lower baseline temperatures and much weaker in those with higher ones. Specifically, as shown in Figure 3 and Column 2 of Table 2, a sustained increase of one additional day per year in the 20–35°C bins raises the stock of latrines after three years by 8.7–10.2 per 1,000 households (a 2.7–3.1% increase), relative to a day in the 15–20°C range. However, in districts with higher baseline temperatures, the effect of higher temperatures is much smaller and only marginally significant (Figure 3 and Column 3 of Table 2), which suggests that people in these districts are better adapted to higher temperatures. The interaction specification in Column 4 of Table 2, which interacts each temperature bin with an indicator for low baseline temperature, confirms that the effects of higher temperatures differ significantly between the two groups for the 20–25 and 30–35°C bins, while the difference for the 25–30°C bin is positive but imprecisely estimated.

I also find that lower temperatures lead to increased latrine investment, particularly in districts with lower baseline temperatures (Figure 3 and Columns 2–3 of Table 2). This positive effect likely reflects that, although these districts are classified as having lower baseline temperatures, they still experience relatively hot climates given the high median cutoff of 25.7°C. Consequently, cold days in these districts can also heighten discomfort, prompting greater investment in latrines. In districts with higher baseline temperatures, the effects of lower temperatures cannot be estimated precisely, likely due to the limited occurrence of days in the lower-temperature bins (as illustrated in Panel A of Figure 1). Accordingly, the interaction terms for the low-temperature bins are not statistically significant (Column 4 of Table 2). The heterogeneity analysis by baseline temperature is therefore informative mainly about the response to high temperatures.

Temperature Deviation Analysis.—One concern with the heterogeneity analysis by baseline temperatures is the use of a fixed reference temperature range of 15–20°C. This approach may not capture the actual variation in discomfort across districts, as it imposes a uniform reference range irrespective of local climatic conditions.

To address this concern, I conduct an alternative analysis that adjusts the reference bin based on each district’s baseline temperatures. This analysis uses standardized temperature deviations from historical means, providing a measure of unusual temperatures that is comparable across districts. Specifically, for each day and district, I calculate how many standard deviations (SDs) a given temperature observation deviates from the historical mean, as follows:

$$TempDev_{d\tau} = \frac{Temp_{d\tau} - \overline{HistTemp_d}}{\sigma_d} \quad (3)$$

where $Temp_{d\tau}$ is the average temperature in district d on day τ , $\overline{HistTemp_d}$ is the historical mean temperature in district d during the pre-study period (2002–2011), and σ_d is the historical standard deviation of temperatures in district d .

I construct the treatment variables by counting the number of days on which $TempDev_{d\tau}$ falls into each of seven SD bins: < -1.5 , $[-1.5, -1]$, $[-1, -0.25]$, $[-0.25, 0.25]$, $[0.25, 1]$, $[1, 1.5]$, > 1.5 .¹⁷ The reference bin is set to the $[-0.25, 0.25]$ SD range, which represents days with temperatures close to the historical average for each district. I then estimate a modified version of regression 1, using the number of days in each SD bin as the treatment variable.¹⁸

The temperature deviation analysis also yields a U-shaped relationship between temperature and latrine investment, in which the estimates tend to be larger for days that depart farther from the district’s historical mean. As shown in Figure 4, a sustained increase of one additional day per year with a temperature more than 1 SD below the historical mean raises the stock of latrines after three years by 5.1–12.1 per 1,000 households, relative to a day in the reference bin.¹⁹ This effect is comparable in magnitude to the results observed for lower temperatures (5–15°C) in the baseline specification. Although the estimates become less precise for higher temperatures, they are mostly positive and largest for days more than 1.5 SD above the historical mean. This pattern suggests that the discomfort from unusual temperatures, whether cold or hot, is greater on days that depart farther from the district’s historical mean, consistent with the discomfort channel. The smaller magnitude for higher than for lower temperatures may nonetheless reflect the income channel, as discussed in the next subsection.

6.2 Mechanism 2: Income Channel

Another mechanism behind the effects of temperature on latrine investment is the income channel. To examine this channel, I conduct heterogeneity analyses by crop area and by SBM policy period. These analyses provide suggestive evidence that the negative effect of the income channel, operating through reduced agricultural output, offsets the positive effect of the discomfort channel, particularly at high temperatures, and that the SBM subsidies relaxed the financial constraints underlying this channel.

Heterogeneity Analysis by Crop Area.—To examine the income channel related to agricultural

¹⁷ The distribution of temperature deviations from the historical mean across the seven SD bins is shown in Appendix Figure C6.

¹⁸ One modification from the baseline specification is that I exclude observations from 2012 to 2014, since their three-year lagged temperature-deviation variables would overlap with the pre-study period (2002–2011) used to define the historical temperature distribution.

¹⁹ All coefficients and standard errors for the results shown in Figure 4 can be found in Appendix Table D5.

production, I estimate heterogeneous effects by crop area.²⁰ Specifically, I compare the effects of temperature in districts with baseline crop areas above the sample median, measured in 2011 (one year prior to the start of the sample period), to those with smaller crop areas. The income channel, which reflects reduced income due to lower agricultural production under extreme temperatures, is expected to be more pronounced in districts with larger crop areas than in those with smaller crop areas. Consequently, districts with larger crop areas are likely to experience smaller (or more negative) effects of temperature on latrine investment, as the negative income channel is expected to more strongly offset the positive effects of the discomfort channel. Moreover, the income channel is expected to be more pronounced for high temperatures, as previous studies have demonstrated that the negative effects of temperature on agricultural production are concentrated in higher temperature ranges in India (Burgess et al., 2017; Colmer, 2021).

Consistent with the income channel, the heterogeneity analysis by crop area shows that the positive effects of higher temperatures are less pronounced in districts with larger crop areas. Both Figure 5 and Columns 1 and 2 of Table 3 show that the effects of higher temperatures on latrine investment are smaller in districts with larger crop areas. In these districts, the effect becomes statistically insignificant above 25°C and turns negative above 30°C. Conversely, districts with smaller crop areas, which are less susceptible to the income channel, show positive effects of higher temperatures that are statistically significant in the 20–35°C bins. The interaction specification in Column 3 of Table 3, which interacts each temperature bin with an indicator for low baseline crop area, is consistent with this interpretation: the interaction terms for all high-temperature bins are positive, suggesting a more positive high-temperature response in districts with smaller crop areas, although each is imprecisely estimated. At lower temperatures, both types of districts show positive effects, consistent with the income channel playing a minimal role in these temperature ranges.²¹ These findings suggest that the negative effect of the income channel, operating through reduced agricultural output, offsets the positive effect of the discomfort channel at high temperatures.

Heterogeneity Analysis by SBM Policy Period.—To further examine the income channel, I estimate heterogeneous effects between the pre- and post-SBM periods, exploiting variation

²⁰ Agricultural data for this analysis are sourced from the ICRISAT (International Crops Research Institute for the Semi-Arid Tropics) District Level Database. I calculate the district-level baseline crop area for 2011 as the total area cultivated with various crops. After matching these data with my main dataset, the sample size is reduced from 614 districts to 528 districts for this heterogeneity analysis.

²¹ The interaction term for the 5–10°C bin is negative and marginally significant, suggesting a larger cold-day response in districts with larger crop areas. As the income channel is expected to operate primarily at high temperatures, this difference is not predicted by the channel and is not central to the analysis.

in the financial cost of latrine construction. As discussed in Section 2.2, the SBM, launched in 2014, provided substantial subsidies that sharply reduced households’ out-of-pocket costs of building a latrine. In the post-SBM period (2014–2019), the subsidy is expected to relax financial constraints on construction, thereby muting the income channel and allowing the positive effects driven by the discomfort channel to manifest more fully. In contrast, in the pre-SBM period (2012–2013), households faced substantially higher out-of-pocket costs, so the income channel is expected to be more binding and to attenuate the positive effects of extreme temperatures on latrine investment.²²

The heterogeneity analysis by SBM policy period suggests that financial constraints attenuate the temperature response and points to an interaction between subsidy policy and temperature-driven adaptation. As shown in Figure 6 and Table 4, extreme temperatures generate positive effects on latrine investment in the post-SBM period in both the high- and low-temperature bins, exhibiting the U-shaped pattern documented in the baseline specification. However, the effects of extreme temperatures in the pre-SBM period are mostly statistically insignificant and economically small. The negative effects observed in the high-temperature bins further suggest that the income channel, operating through reduced agricultural output, may be substantial in the absence of subsidies. The interaction specification in Column 3 of Table 4, which interacts each temperature bin with a post-SBM indicator, confirms that these differences in effects across SBM policy periods are statistically significant in both tails of the temperature distribution. These results are consistent with the income channel attenuating the discomfort response when households bear a larger share of the construction cost. More broadly, these results suggest a complementarity between subsidy policy and climate adaptation: the SBM program enables households to respond to extreme temperature shocks with adaptive investments that would otherwise be limited by financial constraints.

6.3 Alternative Explanations

I explore two alternative explanations, construction feasibility and government relief interventions, but do not find evidence supporting either explanation.

Construction Feasibility.—Extreme temperatures could negatively affect latrine investment in the short run by causing delays and increasing construction costs. Because this channel should primarily generate contemporaneous effects, these effects would be expected to be

²² In this analysis, the policy period is assigned by the outcome year, so the estimates capture whether the availability of the subsidy at the time of latrine construction changes the response to contemporaneous and recent temperature shocks.

negative if it were dominant. Instead, the contemporaneous coefficients in the distributed lag model are positive in every temperature bin (Panel B of Figure 2 and Appendix Table D2), so construction feasibility cannot be the dominant mechanism.

Government Relief.—Another possible explanation is government intervention through relief efforts, whereby latrines may be constructed in response to extreme weather events such as heat waves or cold waves. However, the Indian government’s heat-wave relief guidelines do not include latrine construction as part of their action plan (NDMA, 2019). More general relief measures may also mitigate income losses caused by extreme temperatures, but such assistance would primarily offset the negative income shock rather than generate an increase in income and is therefore unlikely to explain the observed increase in latrine investment. Thus, there is limited evidence supporting the relevance of this channel.

7 Intensive-Margin Effect on Latrine Use

Extreme temperatures can affect not only latrine investment but also the extent of latrine use after construction at the intensive margin, as the discomfort and income channels may similarly operate at this margin. To examine this effect, I use a household-level panel dataset that captures latrine use over time, exploiting temperature variation at the village level. I find that, once latrines are constructed, extreme temperatures generally do not affect the extent of latrine use at the intensive margin, except at very high temperatures.

7.1 Empirical Strategy

I exploit presumably random variation in village-level temperature across the two survey rounds of the Rural Sanitation Surveys to examine the effect of temperature on latrine use. Specifically, I adopt a fixed effects specification that follows the same approach as regression 1.

$$LatrineUse_{hvtm} = \sum_j \beta_j^{USE} BinTemp_{vtmj} + \sum_k \delta_k^{USE} DecileRain_{vtmk} + \eta_v + \nu_t + \theta_m + \varepsilon_{hvtm} \quad (4)$$

where h indexes households, v indexes villages, t indexes the two Rural Sanitation Surveys rounds in 2013–2014 and 2018, m indexes the survey months, j indexes temperature bins, and k indexes rainfall bins. $LatrineUse_{hvtm}$ represents the latrine use rate of household h in village v in survey round t and month m . It is defined as the proportion of household members using latrines out of the total number of members in that household. $BinTemp_{vtmj}$

is the number of days in which the average temperature is in temperature bin j . I categorize temperature into the same eight bins and use the same 15–20°C reference bin as in the latrine investment specification, although temperature here varies at the village level rather than the district level. As in the latrine investment specification, I control for the number of days in each rainfall decile bin ($DecileRain_{vtmk}$). I include village fixed effects (η_v) to exploit the presumably random variation in village-level temperature across the two survey rounds and to control for time-invariant village-level unobservables that can affect latrine use. I also include survey-round fixed effects (ν_t) to control for the trend in latrine use (e.g., an increase in latrine use because of extensive promotion under the SBM). Survey-month fixed effects (θ_m) are included to control for seasonality in latrine use, which may be driven by within-year weather fluctuations. Standard errors are clustered at the village level because the temperature variation is observed at that level.

The coefficients of interest are β_j^{USE} , which capture the effect of an additional day in temperature bin j on the latrine use rate, relative to a day in the 15–20°C bin. The temperature effect is identified from village-specific deviations in temperature exposure across the two survey rounds, net of survey-round and survey-month fixed effects, and the presumably random nature of these deviations allows β_j^{USE} to be interpreted as causal.

In this analysis, I examine the effect of temperature on latrine use conditional on latrine ownership, thereby isolating the intensive-margin effect from the influence of latrine investment decisions. To do so, I restrict the sample to households that owned latrines in both survey rounds. Specifically, the baseline specification focuses on 437 households in 107 villages from the total sample of 1,188 households.²³

The treatment variable, $BinTemp_{vtmj}$, is constructed by counting the number of days that fall into temperature bin j within a specified reference period leading up to the survey date of household h . The Rural Sanitation Surveys were conducted over multiple months, resulting in variation in survey dates across households.²⁴ Specifically, I construct the treatment variable using daily temperature data from X days before the survey date up to the day before the survey. The reference periods, X , are 1 week, 2 weeks, 1 month, 3 months, 6 months, and 12 months. The rainfall control, $DecileRain_{vtmk}$, is constructed analogously.

The analysis of latrine use focuses on short-run responses to temperature rather than on persistent effects, as latrine use can vary from day to day. The outcome variable is based on each household member’s self-reported usual practice of latrine use. Given the potential for recall bias, however, respondents may base their reports on more recent behaviors, influenced

²³ As a robustness check, I also provide results on latrine use without conditioning on latrine ownership, using the entire sample.

²⁴ The first-round survey was conducted between November 2013 and December 2014, and the second-round survey was conducted between August and December 2018 for households included in the final sample.

by recent temperature fluctuations. Thus, I measure temperature exposure over shorter reference periods of 1 week, 2 weeks, and 1 month, for which I expect the temperature effects to be more pronounced. I also measure temperature exposure over longer reference periods of 3, 6, and 12 months to examine whether the effects persist or weaken as the reference period expands.

7.2 Results

I find that temperature does not influence latrine use at the intensive margin across most temperature ranges. However, when temperature exposure is measured over one week to one month, extremely hot days increase latrine use in the short run, suggesting that the discomfort channel also operates at the intensive margin.

In Figure 7, I do not find a significant effect of temperature on latrine use rates conditional on latrine ownership across most temperature bins, regardless of the reference period used.²⁵ This result, coupled with findings on latrine investment, shows that while extreme temperatures cumulatively increase latrine investment, they generally do not affect latrine use at the intensive margin after construction.

Several reasons may explain the limited effect at the intensive margin. First, the baseline rate of latrine use, conditional on ownership, is already high: on average, 79% of household members use latrines if the household owns one in the first survey round (as shown in Table 1). With such a high baseline, the potential for further increases in use due to extreme temperatures is limited. Second, this result is specific to the four northern Indian states included in the Rural Sanitation Surveys dataset, which are known for their relatively hot climates. As discussed in Section 6.1, people in these states are likely to be better adapted to high temperatures, implying that hot days may have a limited impact on latrine use except during extreme heat events.²⁶ At the same time, the hot climate makes it difficult to estimate the effects of cold days. The dataset contains no observations in the lower-temperature bins (below 5°C and 5–10°C) across most reference periods, except for the 5–10°C bin for the 12-month reference period (Panel B of Figure 1). Consequently, I cannot effectively test the effects of cold days, even though such effects may be particularly pronounced in these generally hot-climate states.

²⁵ All coefficients and standard errors for the results shown in Figure 7 can be found in Appendix Table D6. Similar findings are observed when analyzing the full sample of households without conditioning on latrine ownership, as shown in Appendix Figure C7.

²⁶ In addition, the discomfort channel may be weaker at the intensive margin on hot days, because the decision to use a latrine depends on the discomfort of open defecation relative to that of using the latrine itself, and latrines can also become hot inside under high temperatures. Hot days may therefore make latrines only modestly more comfortable than open defecation, which is consistent with the limited temperature effects.

However, I find that extremely hot days increase latrine use in the short run. Figure 7 shows the positive effects of temperatures above 35°C when adopting reference periods of 1 week, 2 weeks, or 1 month. Specifically, an additional hot day with an average temperature above 35°C increases the latrine use rate by 0.15–0.49 (i.e., 15–49 percentage points), or 19–62% of the baseline use rate, relative to a day in the 15–20°C range. This result suggests that discomfort from extreme heat also plays a role in driving increased latrine use at the intensive margin after construction.²⁷

8 Conclusion

I document that extreme temperatures have a positive, persistent effect on latrine investment. My analysis suggests that the main underlying mechanism is the discomfort channel, whereby households construct latrines to avoid the greater discomfort of walking outside for open defecation under extreme temperatures. This adaptive response, partly attenuated by the income channel, emerges more strongly when construction subsidies under the SBM relax households’ financial constraints. The resulting adaptive investment can reduce open defecation and ultimately improve human health by reducing diarrheal child mortality.

My results point to a potential benefit of the increased occurrence of extreme weather under climate change, which most past studies, focusing on its negative consequences, have not documented. Moreover, I document a new mechanism behind the persistent (rather than short-run) effects of temperature: adaptive investment in durable health technologies that remain in place and can be used over multiple years.

A back-of-the-envelope calculation suggests health gains comparable in magnitude to the health damages estimated in previous studies. My results suggest that extreme temperatures can reduce diarrheal child mortality by increasing latrine investment. Thus, I calculate this positive health effect by multiplying the effect of temperature on latrine investment estimated in this paper by the effect of latrine construction on the diarrheal child mortality rate in rural India reported in Motohashi (2026).²⁸ This back-of-the-envelope calculation shows that a sustained increase of one additional cold or hot day per year could reduce the diarrheal post-neonatal mortality rate by 0.13–0.73%.²⁹ The magnitude of this positive

²⁷ Conversely, the income channel is expected to have limited short-run effects for up to one month, as the costs associated with latrine use, primarily latrine emptying costs, are typically incurred only every few years under usual emptying cycles.

²⁸ Specifically, I use the estimates from Column 1 of Table 2 in this paper, along with the estimated effect from Motohashi (2026), which shows that one additional upstream latrine per square kilometer reduces the diarrheal post-neonatal mortality rate by 0.43% from the pre-SBM period. More detailed steps are described in Appendix B.

²⁹ More concretely, I find that one additional day per year with an average temperature of below 5°C, 5–10°C, 10–15°C, 20–25°C, and 25–30°C could decrease the diarrheal post-neonatal mortality rate by 0.73%,

effect is comparable to estimates of the negative health impacts of extreme temperatures in the previous literature. For example, Burgess et al. (2017) report that an additional hot day increased all-age mortality in rural India by 0.21–0.47% during 1957–2000, before widespread sanitation investments. This comparison suggests that the health impact of extreme temperatures is more nuanced than previously thought, as the direct damages may be partly offset by adaptive health investments.

My results have two broad implications for climate change policies and health behaviors in developing countries. First, adaptation to greater variability in temperature under climate change might have unintended positive consequences. Under extreme temperatures, people can shift from outdoor behaviors that are harmful to human health (e.g., open defecation) to health-improving behaviors (e.g., latrine investment) that are conducted indoors. This adaptive response can be strengthened by subsidy policies that lower the cost of adopting these technologies, pointing to a complementarity between subsidy policy and climate adaptation.

Second, my findings on temperature-induced adaptive investments in sanitation technologies have implications for a range of other outdoor health behaviors in developing countries. For example, under extreme temperatures, people may shift from collecting and using biomass for cooking to using cleaner fuels such as liquefied petroleum gas, or from collecting unsafe spring water to using safe tap water, thereby avoiding these harmful outdoor behaviors. However, extreme temperatures may also discourage some beneficial outdoor behaviors, such as traveling to formal health facilities outside the village rather than relying on informal providers within it, and thereby entail health costs. Investigating the potential health benefits and costs of extreme temperatures in different settings may be a fruitful area for future research.

0.59%, 0.13%, 0.22%, and 0.16%, respectively.

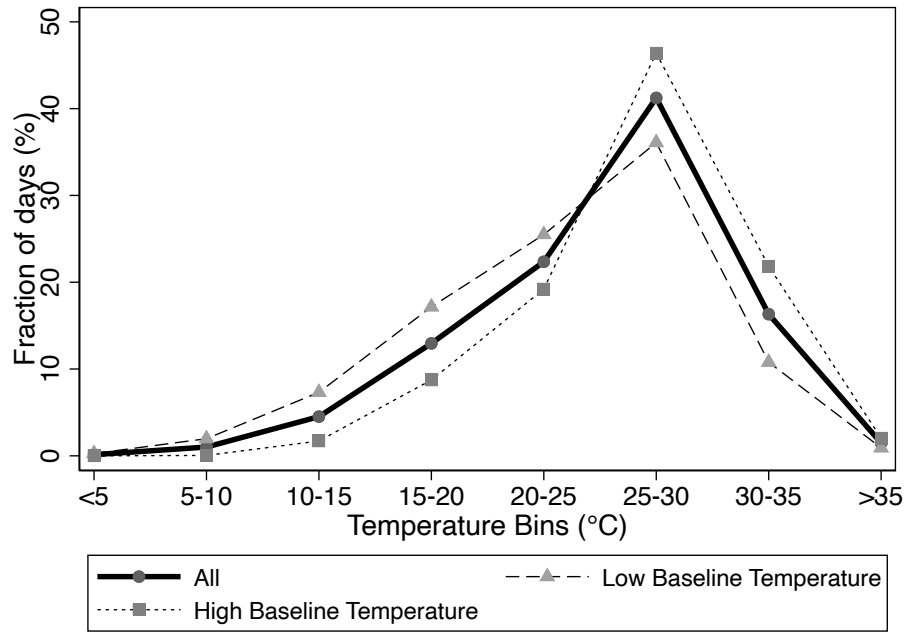
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Panel A. Latrine Investment Analysis



Panel B. Latrine Use Analysis

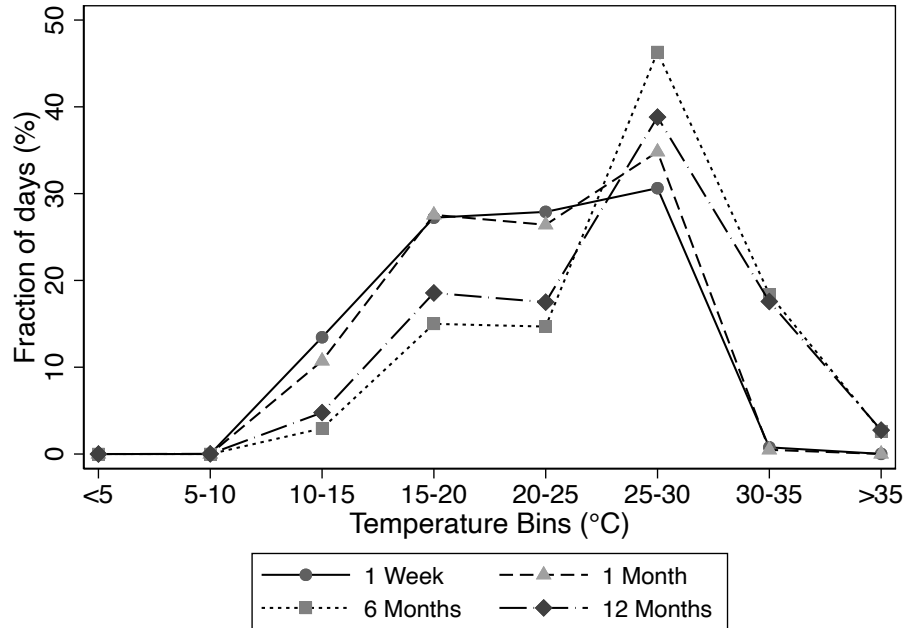
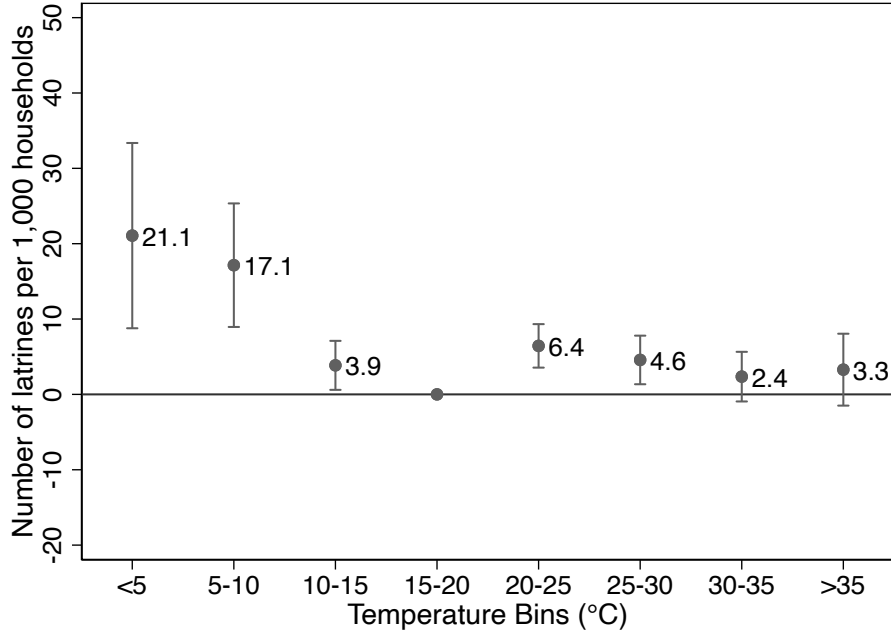


Figure 1: Daily Temperature Distributions

Notes: This figure reports the distributions of daily average temperatures used in the latrine investment analysis (Panel A) and the latrine use analysis (Panel B). Panel A reports distributions for (i) all districts, (ii) districts with baseline temperatures lower than the sample median, and (iii) districts with higher baseline temperatures, using daily temperatures at the district level across India from 2012 to 2019. Panel B reports distributions for four representative reference periods (1 week, 1 month, 6 months, and 12 months), using daily temperatures at the village level in the Rural Sanitation Surveys sample over two survey rounds in 2013–2014 and 2018.

Panel A. Cumulative Effect over a Three-Year Period



Panel B. Decomposed Contemporaneous and Lagged Effects

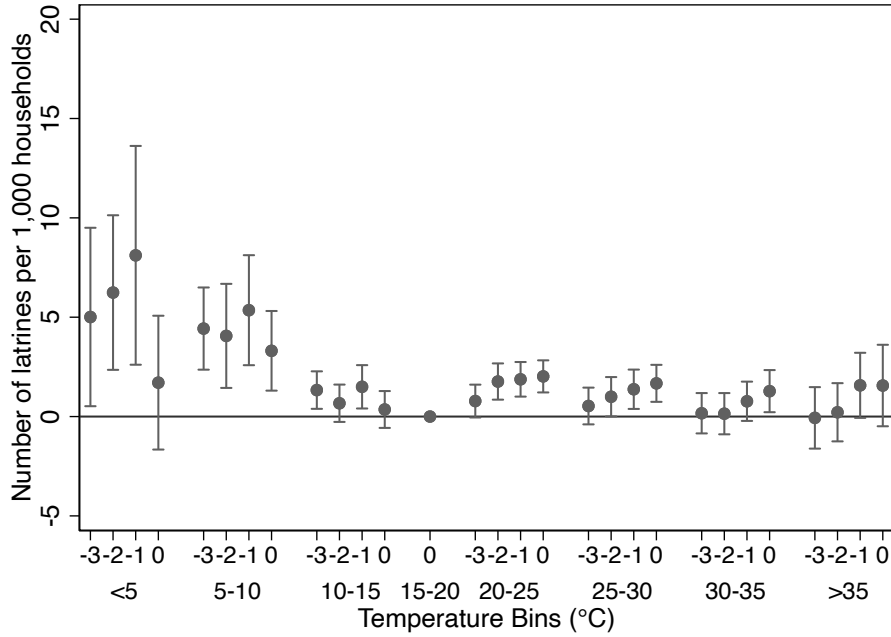


Figure 2: Effect of Temperature on Latrine Investment

Notes: This figure plots the estimated effects of temperature on latrine investment. Panel A reports the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperatures. Panel B reports all estimates of the contemporaneous effect (shown at lag 0) and the lagged effects (shown at lags -1 to -3). The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each rainfall decile bin, with up to three years of lags. The $15\text{--}20^\circ\text{C}$ bin serves as a reference bin and is dropped from the regression. Markers with whisker lines plot temperature bin estimates and associated 95% confidence intervals. Standard errors are clustered at the district level.

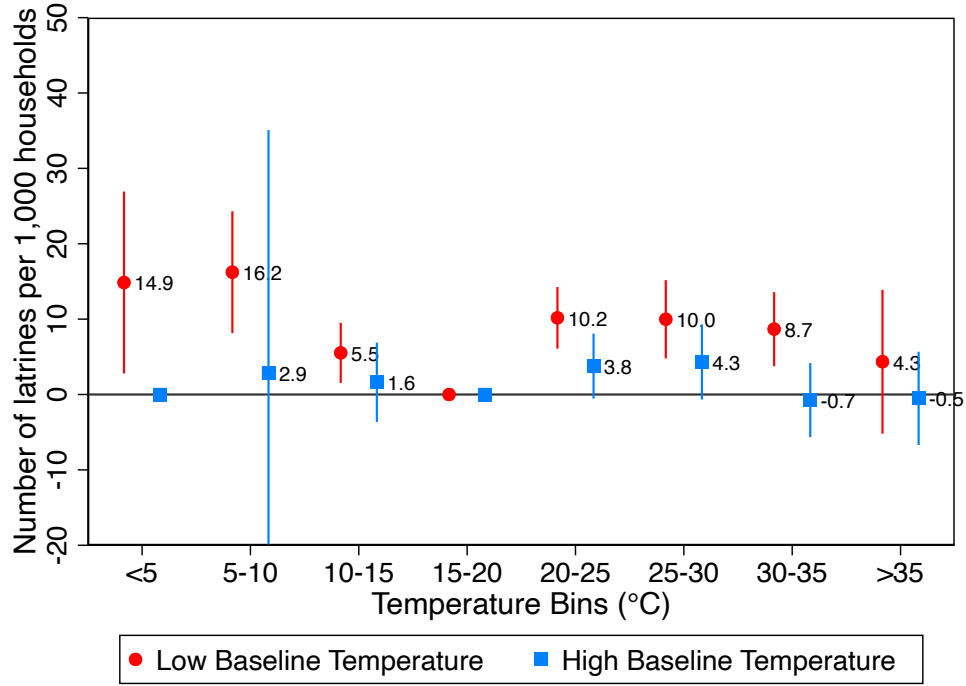


Figure 3: Heterogeneous Effects of Temperature on Latrine Investment by Baseline Temperature

Notes: This figure plots the estimated effects of temperature on latrine investment in districts with baseline temperatures lower than the sample median and in districts with higher baseline temperatures. It reports the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperatures. The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each rainfall decile bin, with up to three years of lags. The 15–20°C bin serves as a reference bin and is dropped from the regression. The markers represent temperature bin estimates, while the lines show the associated 95% confidence intervals, which are truncated at the minimum y-axis value. Standard errors are clustered at the district level.

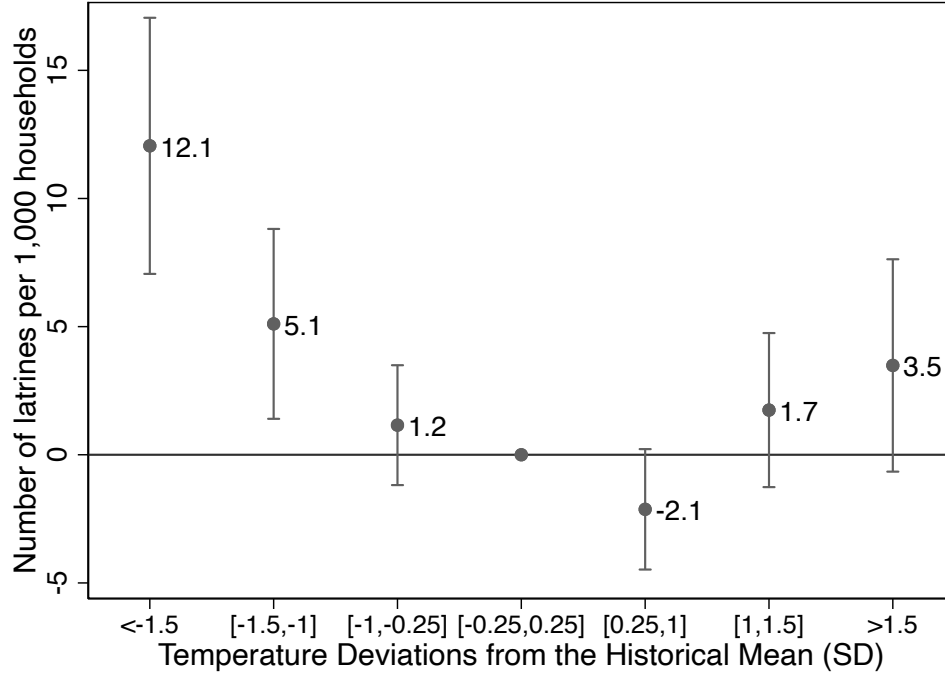


Figure 4: Effect of Temperature Deviations from the Historical Mean on Latrine Investment

Notes: This figure plots the estimated effects of temperature deviations from the historical mean, defined over 2002–2011, on latrine investment. It reports the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperature deviations. The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each rainfall decile bin, with up to three years of lags. The sample is limited to 2015–2019 to avoid overlap between lagged temperature-deviation variables and the pre-study period used to define the historical temperature distribution. The $[-0.25, 0.25]$ standard deviation bin serves as a reference bin and is dropped from the regression. Markers with whisker lines plot temperature-deviation bin estimates and associated 95% confidence intervals. Standard errors are clustered at the district level.

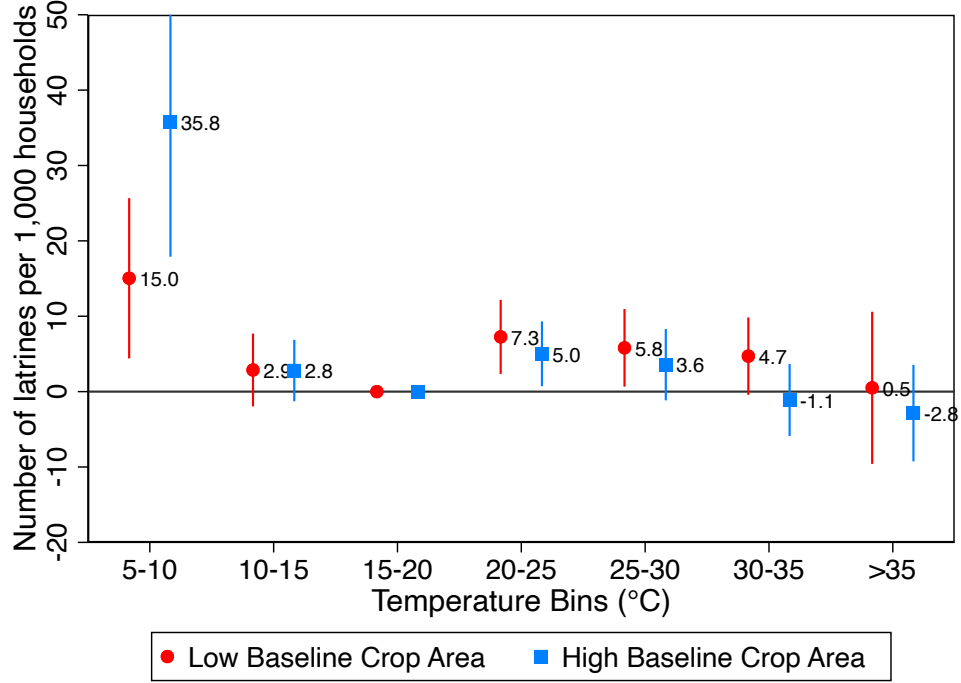


Figure 5: Heterogeneous Effects of Temperature on Latrine Investment by Baseline Crop Area

Notes: This figure plots the estimated effects of temperature on latrine investment in districts with baseline crop areas lower than the sample median and in districts with higher crop areas. It reports the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperatures. The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each rainfall decile bin, with up to three years of lags. The sample is limited to districts with baseline crop area data. The 15–20°C bin serves as a reference bin and is dropped from the regression. The coefficient for the below 5°C bin, which is estimated only for districts with lower baseline crop areas, is omitted from this figure due to its large value; it is reported in Table 3. The markers represent temperature bin estimates, while the lines show the associated 95% confidence intervals, which are truncated at the maximum y-axis value. Standard errors are clustered at the district level.

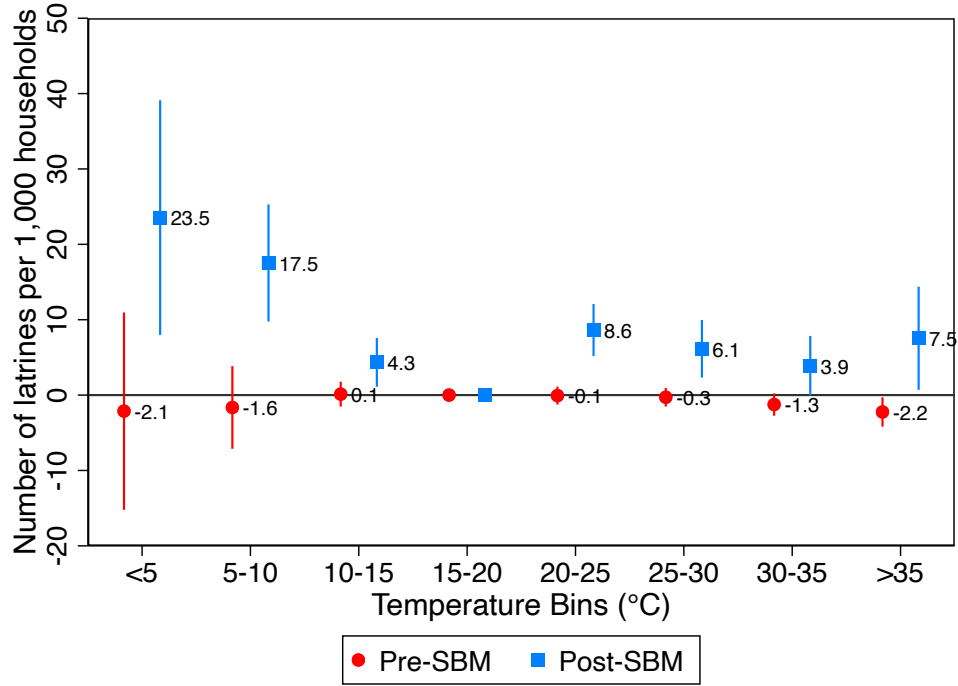


Figure 6: Heterogeneous Effects of Temperature on Latrine Investment by SBM Policy Period

Notes: This figure plots the estimated effects of temperature on latrine investment in the pre-SBM period (2012–2013) and the post-SBM period (2014–2019). It reports the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperatures. The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each rainfall decile bin, with up to three years of lags. The 15–20°C bin serves as a reference bin and is dropped from the regression. The markers represent temperature bin estimates, while the lines show the associated 95% confidence intervals, which are truncated at the maximum y-axis value. Standard errors are clustered at the district level.

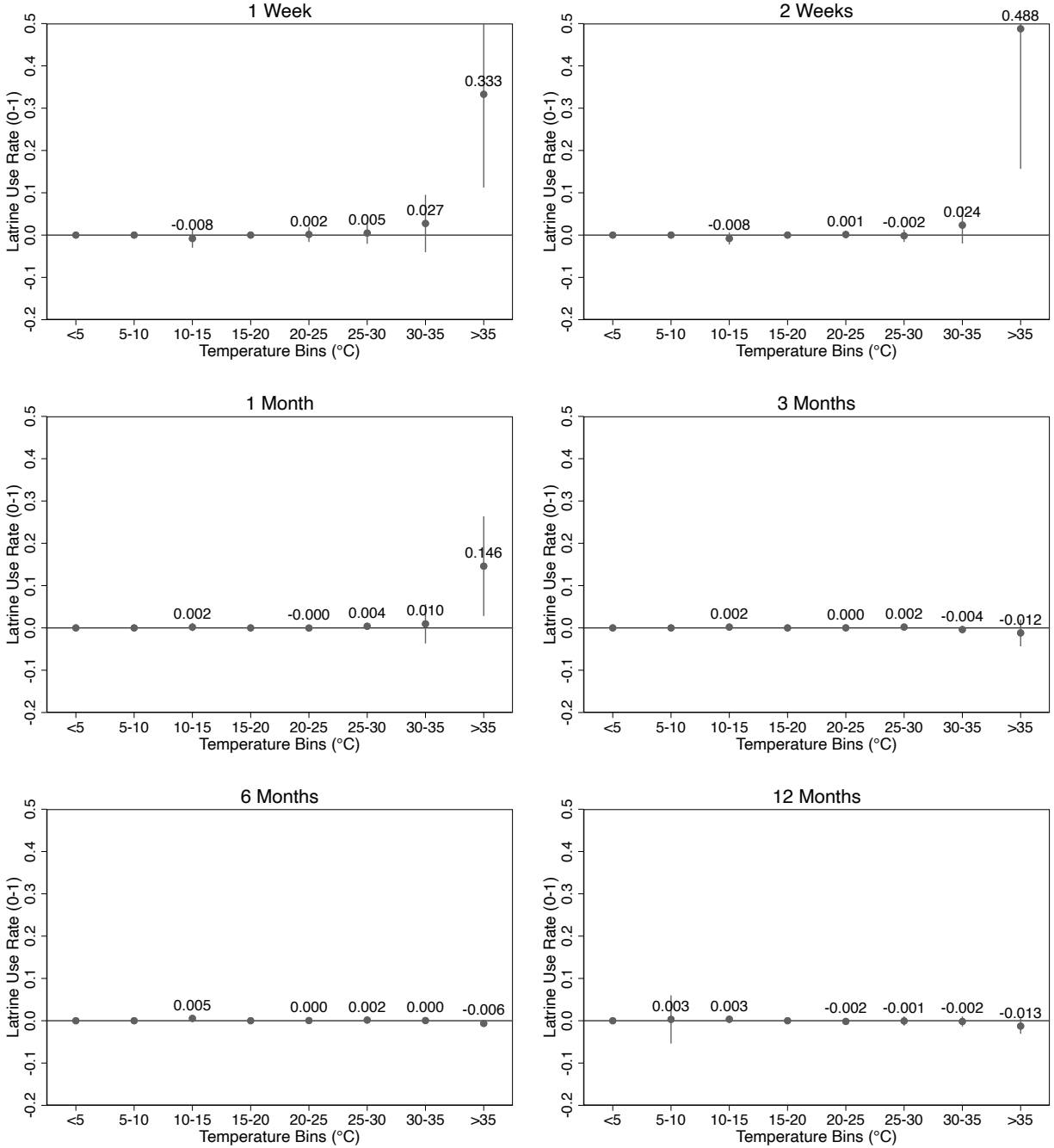


Figure 7: Effect of Temperature on Latrine Use (Conditional on Ownership)

Notes: This figure plots the estimated effects of temperature on latrine use rates for households that owned latrines in both survey rounds, with temperature exposure measured over different reference periods. Each specification includes village fixed effects, survey-round fixed effects, survey-month fixed effects, and controls for the number of days in each rainfall decile bin over the same reference period as the temperature variables. The 15–20°C bin serves as a reference bin and is dropped from the regression. The markers represent temperature bin estimates, while the lines show the associated 95% confidence intervals, which are truncated at the maximum y-axis value. Temperature bins in which no days are observed for the corresponding reference period (the below 5°C bin in all panels and the 5–10°C bin except in the 12-month panel) cannot be estimated; the markers at zero for these bins are placeholders, not estimates. Standard errors are clustered at the village level.

Table 1: Summary Statistics

	Mean	SD	Min	Max	Observations
<i>Panel A. District-level Latrine Investment (2012-2019)</i>					
Cumulative number of latrines (thousand)	161.64	159.89	0	1468.74	4912
Cumulative number of latrines per 1,000 households	453.51	263.61	0	2081.31	4912
<i>Panel B. Household-level Latrine Use (2013-2014, 2018)</i>					
Latrine use rate 2013-2014 (0-1)	0.32	0.43	0	1	1188
Latrine use rate 2018 (0-1)	0.6	0.45	0	1	1188
Latrine use rate conditional on ownership 2013-2014 (0-1)	0.79	0.32	0	1	437
Latrine use rate conditional on ownership 2018 (0-1)	0.91	0.22	0	1	437
<i>Panel C. District-level Daily Average Temperature (2012-2019)</i>					
Number of days below 5°C per year	0.49	3.79	0	57	4912
Number of days between 5-10°C per year	3.7	13.4	0	92	4912
Number of days between 10-15°C per year	16.43	22.51	0	98	4912
Number of days between 15-20°C per year	47.35	29.73	0	109	4912
Number of days between 20-25°C per year	81.7	40.68	0	316	4912
Number of days between 25-30°C per year	150.55	48.87	8	364	4912
Number of days between 30-35°C per year	59.61	41.27	0	192	4912
Number of days above 35°C per year	5.43	8.44	0	97	4912
Number of days with temperatures >1.5 SD below historical mean	34.01	14.61	0	67	4912
Number of days with temperatures 1-1.5 SD below historical mean	40.09	11.63	3	73	4912
Number of days with temperatures 0.25-1 SD below historical mean	60.97	17.7	26	149	4912
Number of days with temperatures within 0.25 SD of historical mean	56.11	22.92	15	124	4912
Number of days with temperatures 0.25-1 SD above historical mean	108.65	32.24	23	170	4912
Number of days with temperatures 1-1.5 SD above historical mean	46.06	16.1	9	98	4912
Number of days with temperatures >1.5 SD above historical mean	19.35	17.94	0	104	4912
<i>Panel D. District-level Baseline Characteristics</i>					
Pre-SBM latrine coverage 2013 (0-1)	0.4	0.25	0	0.99	614
Baseline (historical) mean temperature 2002-2011 (°C)	25.11	2.29	16.01	29.14	614
Baseline crop area 2011 (thousand ha)	356.07	305.17	0.41	1845.11	528

Notes: Panel A reports summary statistics of district-level variables on latrine investment. Panel B reports summary statistics of household-level variables on latrine use in each round of the Rural Sanitation Surveys. Panel C reports summary statistics on the distribution of daily average temperature at the district level. Panel D reports summary statistics of the baseline district characteristics used in the heterogeneity analyses.

Table 2: Cumulative Effects of Temperature on Latrine Investment: All Districts and Heterogeneity by Baseline Temperature (Number of Latrines per 1,000 Households)

	All	Baseline Temperature		Interaction
	(1)	(2)	(3)	(4)
	All	Low	High	Interaction
Number of days below 5°C	21.071*** (6.273)	14.853** (6.133)	- -	14.853** (6.093)
Number of days below 5°C × Low Baseline Temperature				- -
Number of days 5-10°C	17.148*** (4.183)	16.226*** (4.105)	2.877 (16.369)	2.877 (16.451)
Number of days 5-10°C × Low Baseline Temperature				13.349 (16.949)
Number of days 10-15°C	3.855** (1.659)	5.521*** (2.023)	1.622 (2.672)	1.622 (2.686)
Number of days 10-15°C × Low Baseline Temperature				3.900 (3.355)
Number of days 20-25°C	6.439*** (1.472)	10.172*** (2.079)	3.765* (2.189)	3.765* (2.200)
Number of days 20-25°C × Low Baseline Temperature				6.406** (3.018)
Number of days 25-30°C	4.572*** (1.645)	9.979*** (2.638)	4.318* (2.529)	4.318* (2.541)
Number of days 25-30°C × Low Baseline Temperature				5.661 (3.651)
Number of days 30-35°C	2.357 (1.681)	8.675*** (2.496)	-0.736 (2.494)	-0.736 (2.507)
Number of days 30-35°C × Low Baseline Temperature				9.412*** (3.526)
Number of days above 35°C	3.282 (2.434)	4.346 (4.848)	-0.513 (3.145)	-0.513 (3.161)
Number of days above 35°C × Low Baseline Temperature				4.859 (5.761)
Observations	4,912	2,456	2,456	4,912
R ²	0.930	0.929	0.936	0.934
Number of Districts	614	307	307	614
Mean of Dep. Variable	266.178	325.021	207.335	266.178

Notes: This table reports estimated effects of temperature on latrine investment. The 15–20°C bin serves as a reference bin and is dropped from the regression. Standard errors, clustered at the district level, are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. All columns report the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperatures. Column 1 reports the estimated effects in all districts. Column 2 reports the estimated effects in districts with baseline temperatures lower than the sample median, while Column 3 reports the estimated effects in districts with higher baseline temperatures. Column 4 reports estimates from a specification that interacts the number of days in each temperature bin with an indicator for low baseline temperature, allowing for a direct test of heterogeneous temperature effects by baseline temperature. Columns 1–3 include district fixed effects and state-by-year fixed effects, and control for the number of days in each rainfall decile bin, with up to three years of lags. Column 4 includes district fixed effects, state-by-year fixed effects, and rainfall-decile controls with up to three years of lags, each interacted with the low-baseline-temperature indicator. The means of dependent variables are calculated for the pre-SBM period (2012–2013). Dashes indicate temperature bins that cannot be estimated because no days fall into the bin for the corresponding sample.

Table 3: Heterogeneous Effects of Temperature on Latrine Investment by Baseline Crop Area
(Number of Latrines per 1,000 Households)

	Baseline Crop Area		Interaction
	(1) Low	(2) High	(3) Interaction
Number of days below 5°C	77.161*** (16.284)	- -	77.161*** (16.178)
Number of days below 5°C × Low Baseline Crop Area			- -
Number of days 5-10°C	15.037*** (5.399)	35.761*** (9.074)	35.761*** (9.118)
Number of days 5-10°C × Low Baseline Crop Area			-20.723* (10.578)
Number of days 10-15°C	2.868 (2.451)	2.803 (2.070)	2.803 (2.080)
Number of days 10-15°C × Low Baseline Crop Area			0.066 (3.202)
Number of days 20-25°C	7.265*** (2.495)	5.019** (2.183)	5.019** (2.194)
Number of days 20-25°C × Low Baseline Crop Area			2.245 (3.310)
Number of days 25-30°C	5.810** (2.614)	3.569 (2.403)	3.569 (2.414)
Number of days 25-30°C × Low Baseline Crop Area			2.241 (3.546)
Number of days 30-35°C	4.707* (2.609)	-1.109 (2.432)	-1.109 (2.444)
Number of days 30-35°C × Low Baseline Crop Area			5.817 (3.562)
Number of days above 35°C	0.506 (5.125)	-2.849 (3.251)	-2.849 (3.266)
Number of days above 35°C × Low Baseline Crop Area			3.355 (6.049)
Observations	2,112	2,112	4,224
R ²	0.945	0.935	0.942
Number of Districts	264	264	528
Mean of Dep. Variable	301.980	224.894	263.437

Notes: This table reports estimated effects of temperature on latrine investment. The 15–20°C bin serves as a reference bin and is dropped from the regression. Standard errors, clustered at the district level, are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. All columns report the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperatures. The sample is limited to districts with baseline crop area data. Column 1 reports the estimated effects in districts with baseline crop areas lower than the sample median, while Column 2 reports the estimated effects in districts with higher crop areas. Column 3 reports estimates from a specification that interacts the number of days in each temperature bin with an indicator for low baseline crop area, allowing for a direct test of heterogeneous temperature effects by baseline crop area. Columns 1–2 include district fixed effects and state-by-year fixed effects, and control for the number of days in each rainfall decile bin, with up to three years of lags. Column 3 includes district fixed effects, state-by-year fixed effects, and rainfall-decile controls with up to three years of lags, each interacted with the low-baseline-crop-area indicator. The means of dependent variables are calculated for the pre-SBM period (2012–2013). Dashes indicate temperature bins that cannot be estimated because no days fall into the bin for the corresponding sample.

Table 4: Heterogeneous Effects of Temperature on Latrine Investment by SBM Policy Period
(Number of Latrines per 1,000 Households)

	SBM Policy Period		Interaction
	(1) Pre-SBM	(2) Post-SBM	(3) Interaction
Number of days below 5°C	-2.127 (6.665)	23.549*** (7.927)	-2.127 (6.597)
Number of days below 5°C × Post-SBM			25.676** (11.399)
Number of days 5-10°C	-1.637 (2.790)	17.529*** (3.958)	-1.637 (2.761)
Number of days 5-10°C × Post-SBM			19.167*** (4.712)
Number of days 10-15°C	0.127 (0.840)	4.342*** (1.648)	0.127 (0.831)
Number of days 10-15°C × Post-SBM			4.215** (1.855)
Number of days 20-25°C	-0.067 (0.604)	8.637*** (1.756)	-0.067 (0.598)
Number of days 20-25°C × Post-SBM			8.705*** (1.887)
Number of days 25-30°C	-0.288 (0.629)	6.143*** (1.941)	-0.288 (0.622)
Number of days 25-30°C × Post-SBM			6.431*** (2.081)
Number of days 30-35°C	-1.250* (0.756)	3.890* (2.017)	-1.250* (0.749)
Number of days 30-35°C × Post-SBM			5.140** (2.176)
Number of days above 35°C	-2.238** (0.991)	7.535** (3.485)	-2.238** (0.981)
Number of days above 35°C × Post-SBM			9.774** (3.870)
Observations	1,228	3,684	4,912
R ²	0.999	0.931	0.951
Number of Districts	614	614	614
Mean of Dep. Variable	266.178	515.949	266.178

Notes: This table reports estimated effects of temperature on latrine investment. The 15–20°C bin serves as a reference bin and is dropped from the regression. Standard errors, clustered at the district level, are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. All columns report the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperatures. Column 1 reports the estimated effects in the pre-SBM period (2012–2013), while Column 2 reports the estimated effects in the post-SBM period (2014–2019). Column 3 reports estimates from a specification that interacts the number of days in each temperature bin with a post-SBM indicator, allowing for a direct test of heterogeneous temperature effects between the pre- and post-SBM periods. Columns 1–2 include district fixed effects and state-by-year fixed effects, and control for the number of days in each rainfall decile bin, with up to three years of lags. Column 3 includes district fixed effects, state-by-year fixed effects, and rainfall-decile controls with up to three years of lags, each interacted with the post-SBM indicator. The means of dependent variables are calculated for the pre-SBM period (2012–2013) in Columns 1 and 3 and for the post-SBM period (2014–2019) in Column 2.

Online Appendix

Extreme Temperatures and Adaptive Health Investment: Evidence from Sanitation Behaviors in India

Kazuki Motohashi

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A Conceptual Framework on the Effects of Temperature on Sanitation Behaviors

I present a simple conceptual framework to show that extreme temperatures can have two opposing effects on latrine investment and use: (i) a positive effect through the discomfort channel and (ii) a negative effect through the income channel. A household decides the extent to which it uses latrines. Suppose that the discomfort of walking outside for open defecation, s , depends on the latrine use rate $l \in [0, 1]$, as well as on ambient temperature $a \in [0, 1]$. In the context of latrine investment, l can also be interpreted as the probability of constructing a latrine. Correspondingly, $1 - l$ is the rate of practicing open defecation. Denote the cost of constructing a latrine by p .³⁰ For a , a value of 1 denotes a physically uninhabitable ambient temperature (either extremely high or extremely low), and a value of 0 denotes the ideal temperature.

Then, the discomfort of walking outside for open defecation can be expressed as $s(a, 1 - l)$. People experience more discomfort under more extreme temperatures: $\frac{\partial s}{\partial a} > 0$. Moreover, people experience more discomfort with a higher rate of practicing open defecation (lower rate of latrine use): $\frac{\partial s}{\partial l} < 0$.

The household derives utility from consuming a composite good x , whose price is normalized to 1, and experiences disutility from the discomfort of walking outside for open defecation, represented by $s(a, 1 - l)$. To derive an explicit comparative-static expression, I assume that utility is additively separable in consumption and discomfort. The household's utility is therefore given by $u(x) - s(a, 1 - l)$, where $u_x > 0$ and $u_{xx} < 0$.

The budget constraint is $I(a) = lp + x$, where income $I(a)$ depends on ambient temperature because extreme temperatures may reduce agricultural productivity. I assume that income decreases as temperature becomes more extreme, such that $\frac{dI}{da} < 0$.³¹

The household's utility maximization problem is therefore

$$\max_l u(x) - s(a, 1 - l) \quad \text{s.t.} \quad I(a) = lp + x.$$

Substituting the budget constraint into the utility function, the problem can equivalently

³⁰ In the context of latrine use, p represents the maintenance costs associated with emptying pits and septic tanks, as discussed in Section 2.1.

³¹ For simplicity, a indexes the extremeness of temperature symmetrically. Allowing the income function to depend on the direction of the deviation, with $\frac{dI}{da} < 0$ for extremely high temperatures and $\frac{dI}{da} \approx 0$ for extremely low temperatures, leaves equation 7 unchanged and implies that the income channel offsets the discomfort channel mainly at high temperatures, consistent with evidence that the agricultural losses from temperature are concentrated at high temperatures in India (Burgess et al., 2017; Colmer, 2021).

be written as

$$\max_l u(I(a) - lp) - s(a, 1 - l).$$

To simplify the notation, define the derivatives of the composite function $s(a, 1 - l)$ as

$$s_l \equiv \frac{\partial s(a, 1 - l)}{\partial l}, \quad s_{ll} \equiv \frac{\partial^2 s(a, 1 - l)}{\partial l^2}, \quad s_{la} \equiv \frac{\partial}{\partial a} \left[\frac{\partial s(a, 1 - l)}{\partial l} \right].$$

Assuming an interior solution, the first-order condition with respect to l is

$$-pu_x - s_l = 0. \tag{5}$$

Rearranging the first-order condition gives

$$\underbrace{p}_{\text{MC}} = - \underbrace{\frac{s_l}{u_x}}_{\text{MB}},$$

which means that the household chooses the latrine use rate to balance the marginal cost of latrine use against its marginal benefit. The marginal benefit arises from the reduction in the discomfort of walking outside for open defecation.

The second-order condition for an interior maximum is

$$p^2 u_{xx} - s_{ll} < 0, \tag{6}$$

which holds, for example, whenever the discomfort function is weakly convex in the rate of open defecation ($s_{ll} \geq 0$), given $u_{xx} < 0$.

To capture the discomfort channel, I assume that $s_{la} < 0$. This condition means that, as ambient temperature becomes more extreme, an increase in latrine use produces a larger marginal reduction in the discomfort associated with open defecation. In other words, extreme temperatures make avoiding open defecation more valuable.

To examine how the optimal latrine use rate responds to extreme temperatures, let the household's optimal choice be denoted by $l = l(a)$. The first-order condition (equation 5) can then be written explicitly as

$$-pu_x(I(a) - pl(a)) - s_l(a, 1 - l(a)) = 0.$$

Differentiating this first-order condition with respect to a gives

$$-pu_{xx} \left(\frac{dI}{da} - p \frac{dl}{da} \right) - s_{la} - s_{ll} \frac{dl}{da} = 0.$$

Rearranging this expression yields

$$\frac{dl}{da} = \frac{1}{s_{ll} - p^2 u_{xx}} \left\{ \underbrace{-s_{la}}_{\text{Discomfort channel} > 0} + \underbrace{\left(-p u_{xx} \frac{dI}{da}\right)}_{\text{Income channel} < 0} \right\}. \quad (7)$$

Equation 7 shows two opposing channels. First, extreme temperatures increase latrine investment and use through the discomfort channel. Because $s_{la} < 0$, the discomfort-channel term satisfies $-s_{la} > 0$.

Second, extreme temperatures reduce latrine investment and use through the income channel. Because $u_{xx} < 0$ and $\frac{dI}{da} < 0$, the income-channel term satisfies $-p u_{xx} \frac{dI}{da} < 0$. The intuition is that a reduction in income increases the marginal utility of consumption, thereby raising the opportunity cost of spending resources on latrine construction or use.

Therefore, the sign of the net effect depends on the relative magnitudes of the two terms in the numerator, while the denominator satisfies $s_{ll} - p^2 u_{xx} > 0$ by the second-order condition (equation 6). The relative strengths of the discomfort and income channels determine whether extreme temperatures increase or decrease latrine investment and use. My empirical analysis examines which of these channels dominates.³²

Finally, the SBM context in Section 2.2 can be embedded in this framework through the cost of latrine construction. Under the SBM, the subsidy reduces the household's out-of-pocket cost of constructing a latrine from p to $p - \tau$, where τ denotes the subsidy amount. Replacing p with $p - \tau$ in equation 7, the income-channel term becomes $-(p - \tau) u_{xx} \frac{dI}{da}$, which shrinks toward zero as the subsidy covers a larger share of the construction cost, whereas the discomfort-channel term ($-s_{la}$) is unaffected. The subsidy therefore mutes the income channel. This yields the prediction, examined in Section 6.2, that the positive effect of extreme temperatures on latrine investment is more pronounced in the post-SBM period (2014–2019) than in the pre-SBM period (2012–2013).

³² This conceptual framework adopts a static model to illustrate the two underlying channels. For simplicity, the persistence of the effect of extreme temperatures on latrine investment is not explicitly modeled. Such persistence can arise because latrines are durable goods that remain in place and can be used over multiple years after construction.

B Back-of-the-Envelope Calculation of the Health Effect

I conduct a back-of-the-envelope calculation of the health effect of extreme temperatures through increased latrine investment by multiplying the effect of temperature on latrine investment estimated in this paper by the effect of latrine construction on the diarrheal child mortality rate in rural India, as reported in Motohashi (2026).

Regarding the effect of temperature on latrine investment, I refer to the statistically significant estimates presented in Column 1 of Table 2. Specifically, these estimates correspond to an increase in the stock of latrines after three years of 21.1, 17.1, 3.9, 6.4, and 4.6 per 1,000 households in response to a sustained increase of one additional day per year with an average temperature below 5°C or in the 5–10°C, 10–15°C, 20–25°C, and 25–30°C bins, respectively. Multiplying these estimates by the average number of households per district (390.27 thousand in my sample of 614 districts in the 2011 Census) and dividing by the average area per district (4,864.14 square kilometers in the same sample) translates them into an increase in the latrine stock of 1.7, 1.4, 0.31, 0.52, and 0.37 per square kilometer, respectively.

Regarding the effect of latrine construction on the diarrheal post-neonatal mortality rate, I refer to the estimate reported in Motohashi (2026): one additional upstream latrine per square kilometer reduces this mortality rate by 0.011 per 1,000 children, corresponding to a 0.43% decrease from the pre-SBM period.

Finally, multiplying these two estimates yields the health effect of extreme temperatures via increased latrine investment. A sustained increase of one additional day per year with an average temperature below 5°C or in the 5–10°C, 10–15°C, 20–25°C, and 25–30°C bins could decrease the diarrheal post-neonatal mortality rate by 0.73%, 0.59%, 0.13%, 0.22%, and 0.16%, respectively.

C Additional Figures

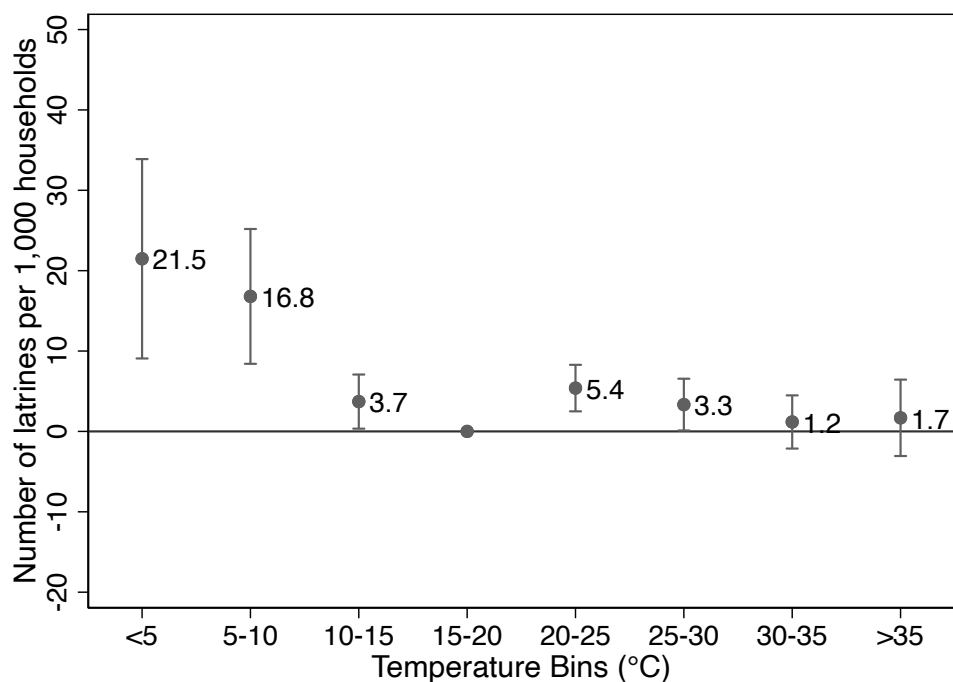
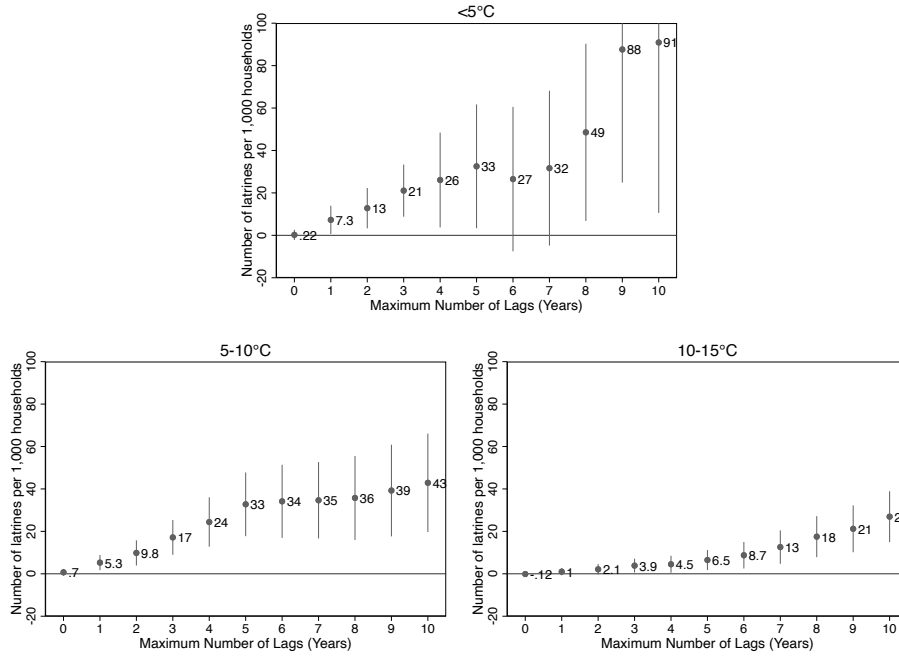


Figure C1: Effect of Temperature on Latrine Investment Excluding Districts with More than 1,000 Latrines per 1,000 Households

Notes: This figure plots the estimated effects of temperature on latrine investment in the sample that excludes the 36 districts in which the cumulative number of latrines per 1,000 households exceeds 1,000 in 2019, where the households recorded by the program outnumber those counted in the 2011 Census (578 districts remain). It reports the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperatures. The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each rainfall decile bin, with up to three years of lags. The 15–20°C bin serves as a reference bin and is dropped from the regression. Markers with whisker lines plot temperature bin estimates and associated 95% confidence intervals. Standard errors are clustered at the district level.

Panel A. Low-Temperature Bins



Panel B. High-Temperature Bins

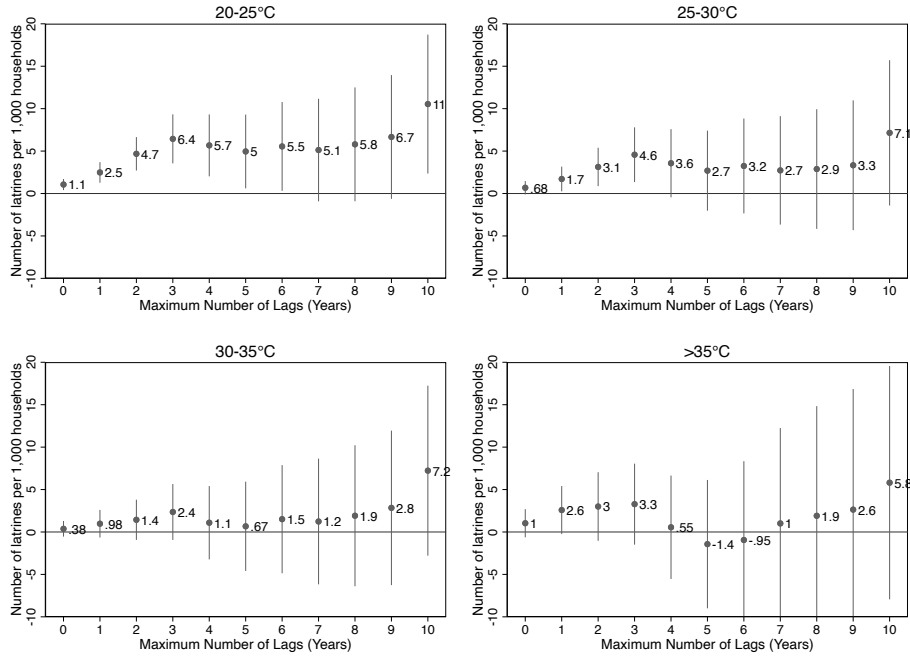


Figure C2: Cumulative Effects of Temperature on Latrine Investment with Different Maximum Numbers of Lags ($\sum_l \beta_{jl}^{INV}$)

Notes: This figure plots the estimated effect of temperature on latrine investment for different temperature bins for different maximum numbers of lags (years). It reports the cumulative effects, representing the sum of contemporaneous and lagged effects. All specifications include district fixed effects and state-by-year fixed effects, and control for the number of days in each rainfall decile bin, using the same maximum number of lags. The 15–20°C bin serves as a reference bin and is dropped from the regression. The markers represent temperature bin estimates, while the lines show the associated 95% confidence intervals, which are truncated at the maximum and minimum y-axis values. Standard errors are clustered at the district level.

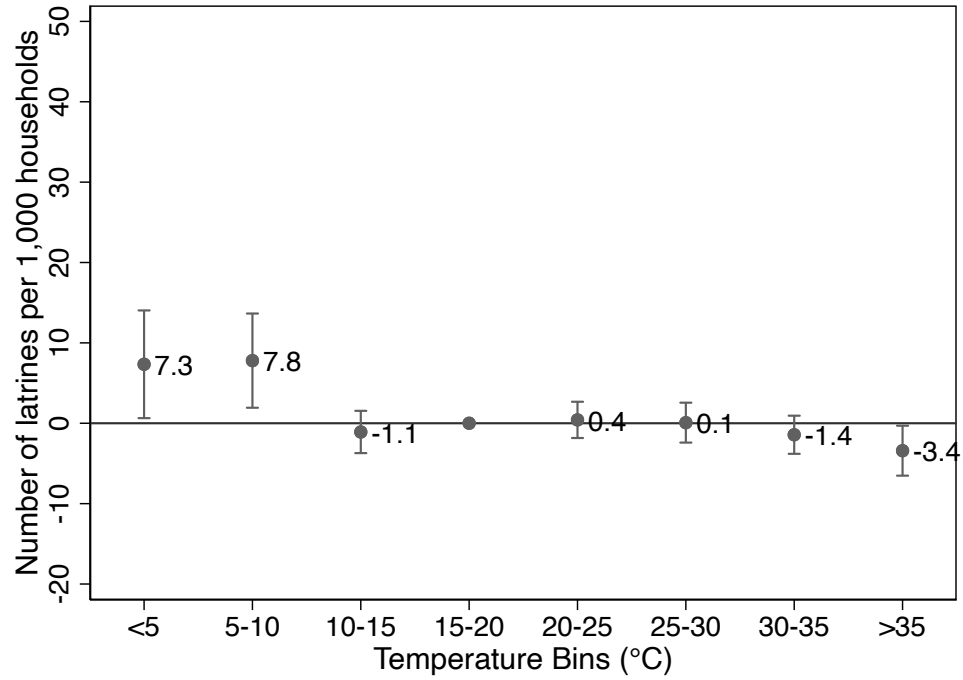


Figure C3: Effect of Temperature on Latrine Investment through 2016

Notes: This figure plots the estimated effects of temperature on latrine investment in the period through 2016 (2012–2016). It reports the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperatures. The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each rainfall decile bin, with up to three years of lags. The 15–20°C bin serves as a reference bin and is dropped from the regression. Markers with whisker lines plot temperature bin estimates and associated 95% confidence intervals. Standard errors are clustered at the district level.

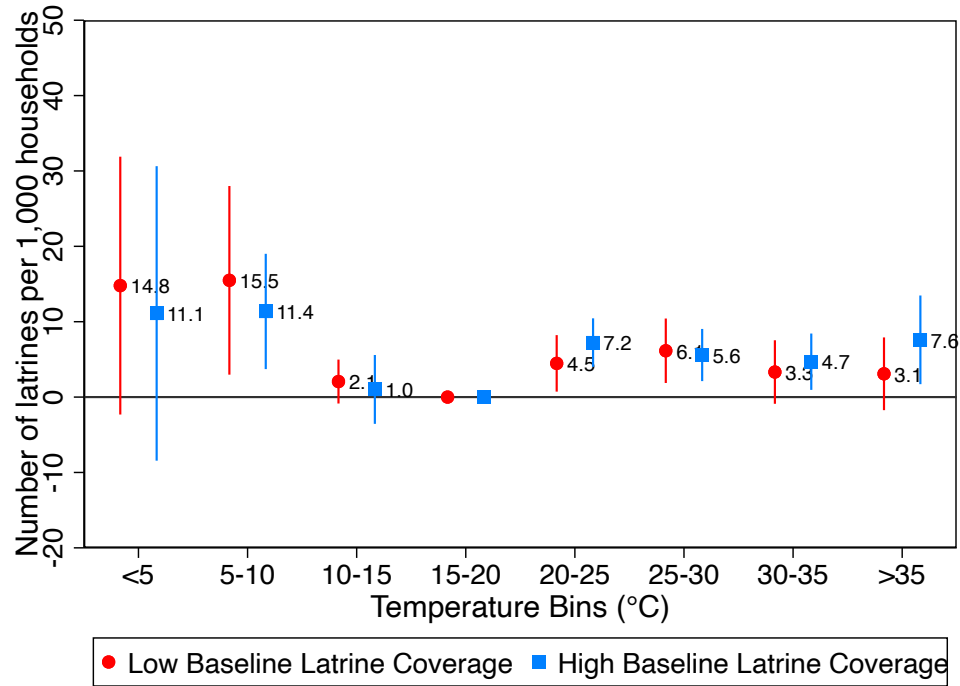


Figure C4: Heterogeneous Effects of Temperature on Latrine Investment by Baseline Latrine Coverage

Notes: This figure plots the estimated effects of temperature on latrine investment in districts with pre-SBM baseline latrine coverage lower than the sample median and in districts with higher baseline coverage. It reports the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged temperatures. The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each rainfall decile bin, with up to three years of lags. The 15–20°C bin serves as a reference bin and is dropped from the regression. The markers represent temperature bin estimates, while the lines show the associated 95% confidence intervals. Standard errors are clustered at the district level.

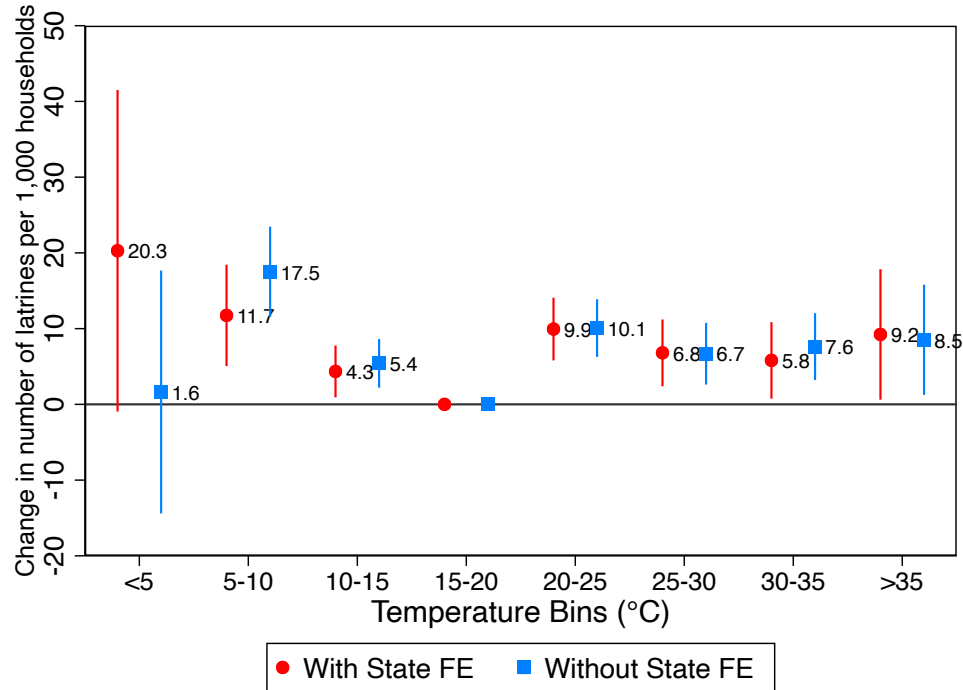


Figure C5: Long-Difference Estimates of the Effect of Temperature on Latrine Investment

Notes: This figure plots the estimated effects of changes in the number of days in each temperature bin on changes in the number of latrines per 1,000 households, where changes are computed for each district as the difference between its mean in the late period (2016–2019) and its mean in the early period (2012–2015). It reports results with and without state fixed effects. The specification controls for changes in the number of days in each rainfall decile bin between the early and late periods. The 15–20°C bin serves as a reference bin and is dropped from the regression. The markers represent temperature bin estimates, while the lines show the associated 95% confidence intervals based on robust standard errors.

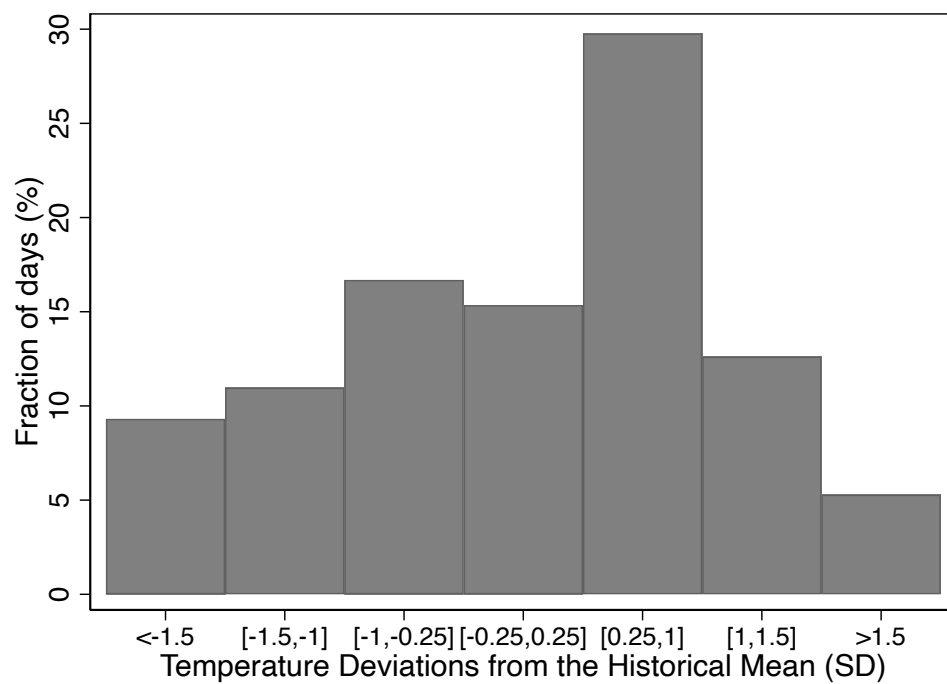


Figure C6: Distribution of Temperature Deviations from the Historical Mean

Notes: This figure reports the distribution of temperature deviations from the historical mean, defined over 2002–2011 and measured in standard deviations, using daily temperatures at the district level across India from 2012 to 2019.

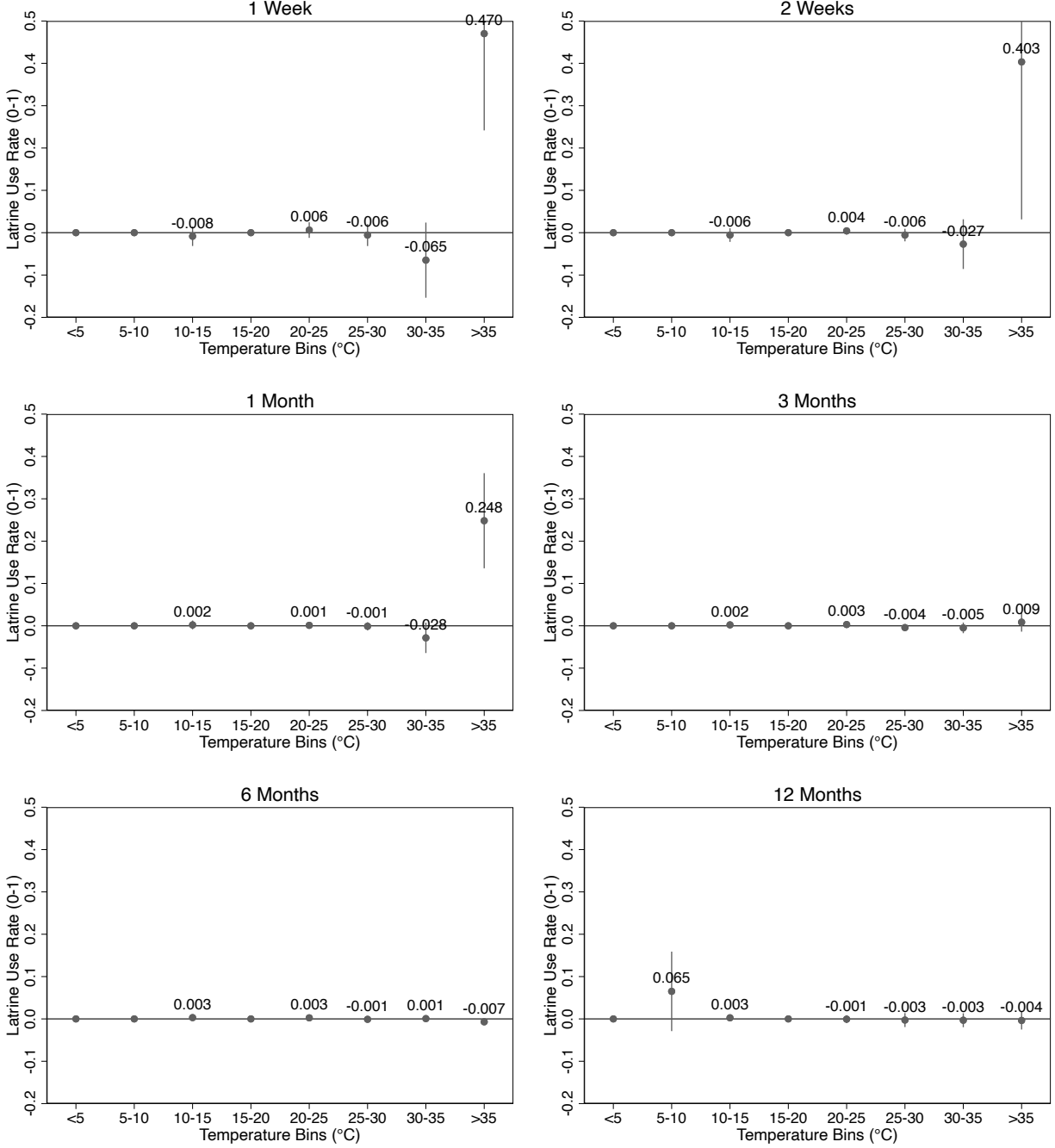


Figure C7: Effect of Temperature on Latrine Use (Unconditional on Ownership)

Notes: This figure plots the estimated effects of temperature on latrine use rates for all households, irrespective of latrine ownership, with temperature exposure measured over different reference periods. Each specification includes village fixed effects, survey-round fixed effects, survey-month fixed effects, and controls for the number of days in each rainfall decile bin over the same reference period as the temperature variables. The 15–20°C bin serves as a reference bin and is dropped from the regression. The markers represent temperature bin estimates, while the lines show the associated 95% confidence intervals, which are truncated at the maximum y-axis value. Temperature bins in which no days are observed for the corresponding reference period (the below 5°C bin in all panels and the 5–10°C bin except in the 12-month panel) cannot be estimated; the markers at zero for these bins are placeholders, not estimates. Standard errors are clustered at the village level.

D Additional Tables

Table D1: Cumulative Effects of Rainfall on Latrine Investment (Number of Latrines per 1,000 Households)

	(1)
Number of days in 6th rainfall decile (0–0.03 mm)	1.089 (0.949)
Number of days in 7th rainfall decile (0.03–0.71 mm)	-1.583** (0.677)
Number of days in 8th rainfall decile (0.71–3.20 mm)	-1.389* (0.807)
Number of days in 9th rainfall decile (3.20–10.11 mm)	-0.063 (0.835)
Number of days in 10th rainfall decile (above 10.11 mm)	0.827 (0.849)
Observations	4,912
R ²	0.930
Number of Districts	614
Mean of Dep. Variable	266.178

Notes: This table reports estimated effects of rainfall on latrine investment, based on the baseline specification in Column 1 of Table 2. The rainfall bins correspond to the sixth through tenth deciles of daily rainfall, and the no-rainfall bin serves as a reference bin and is dropped from the regression. Standard errors, clustered at the district level, are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The column reports the cumulative effects, representing the sum of contemporaneous and lagged effects, when including up to three years of lagged rainfall. The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each temperature bin, with up to three years of lags. The mean of the dependent variable is calculated for the pre-SBM period (2012–2013).

Table D2: Contemporaneous and Lagged Effects of Temperature on Latrine Investment
(Number of Latrines per 1,000 Households)

Low-Temperature Bins		High-Temperature Bins	
Lag 0: Number of days below 5°C	1.707 (1.715)	Lag 0: Number of days 20-25°C	2.022*** (0.411)
Lag 1: Number of days below 5°C	8.113*** (2.803)	Lag 1: Number of days 20-25°C	1.875*** (0.443)
Lag 2: Number of days below 5°C	6.241*** (1.981)	Lag 2: Number of days 20-25°C	1.762*** (0.465)
Lag 3: Number of days below 5°C	5.011** (2.286)	Lag 3: Number of days 20-25°C	0.781* (0.421)
Lag 0: Number of days 5-10°C	3.308*** (1.021)	Lag 0: Number of days 25-30°C	1.671*** (0.474)
Lag 1: Number of days 5-10°C	5.353*** (1.409)	Lag 1: Number of days 25-30°C	1.373*** (0.504)
Lag 2: Number of days 5-10°C	4.059*** (1.333)	Lag 2: Number of days 25-30°C	0.997** (0.504)
Lag 3: Number of days 5-10°C	4.427*** (1.053)	Lag 3: Number of days 25-30°C	0.532 (0.471)
Lag 0: Number of days 10-15°C	0.356 (0.472)	Lag 0: Number of days 30-35°C	1.279** (0.540)
Lag 1: Number of days 10-15°C	1.497*** (0.555)	Lag 1: Number of days 30-35°C	0.768 (0.503)
Lag 2: Number of days 10-15°C	0.669 (0.478)	Lag 2: Number of days 30-35°C	0.144 (0.529)
Lag 3: Number of days 10-15°C	1.332*** (0.481)	Lag 3: Number of days 30-35°C	0.166 (0.517)
Observations	4,912	Lag 0: Number of days above 35°C	1.562 (1.044)
R ²	0.930	Lag 1: Number of days above 35°C	1.571* (0.834)
Number of Districts	614	Lag 2: Number of days above 35°C	0.216 (0.746)
Mean of Dep. Variable	266.178	Lag 3: Number of days above 35°C	-0.066 (0.788)
		Observations	4,912
		R ²	0.930
		Number of Districts	614
		Mean of Dep. Variable	266.178

Notes: This table reports estimated contemporaneous and lagged effects of temperature on latrine investment, when up to three years of lagged temperatures are included. Estimates in both panels are obtained from a single regression that includes all temperature bins. The 15–20°C bin serves as a reference bin and is dropped from the regression. Standard errors, clustered at the district level, are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each rainfall decile bin, with up to three years of lags. The means of dependent variables are calculated for the pre-SBM period (2012–2013).

Table D3: Cumulative Effects of Temperature on Latrine Investment with Different Maximum Numbers of Lags (Number of Latrines per 1,000 Households)

	Maximum Number of Lags (Years)				
	(1) 1 Year	(2) 3 Years	(3) 6 Years	(4) 8 Years	(5) 10 Years
Number of days below 5°C	7.278** (3.393)	21.071*** (6.273)	26.504 (17.361)	48.541** (21.314)	90.909** (40.989)
Number of days 5-10°C	5.271*** (1.823)	17.148*** (4.183)	34.156*** (8.783)	35.762*** (10.104)	42.901*** (11.833)
Number of days 10-15°C	0.999 (0.860)	3.855** (1.659)	8.748*** (3.187)	17.516*** (4.924)	26.940*** (6.120)
Number of days 20-25°C	2.484*** (0.618)	6.439*** (1.472)	5.548** (2.668)	5.795* (3.424)	10.540** (4.176)
Number of days 25-30°C	1.711** (0.743)	4.572*** (1.645)	3.244 (2.854)	2.885 (3.598)	7.149 (4.369)
Number of days 30-35°C	0.976 (0.829)	2.357 (1.681)	1.508 (3.249)	1.916 (4.239)	7.229 (5.110)
Number of days above 35°C	2.582* (1.443)	3.282 (2.434)	-0.947 (4.739)	1.908 (6.594)	5.810 (7.015)
Observations	4,912	4,912	4,912	4,912	4,912
R ²	0.927	0.930	0.931	0.932	0.934
Number of Districts	614	614	614	614	614
Mean of Dep. Variable	266.178	266.178	266.178	266.178	266.178

Notes: This table reports the estimated effects of temperature on latrine investment for different maximum numbers of lags, measured in years. The 15–20°C bin serves as a reference bin and is dropped from the regression. Standard errors, clustered at the district level, are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. All columns report the cumulative effects, representing the sum of contemporaneous and lagged effects. All columns include district fixed effects and state-by-year fixed effects, and control for the number of days in each rainfall decile bin, using the same maximum number of lags. The means of dependent variables are calculated for the pre-SBM period (2012–2013).

Table D4: Long-Difference Estimates of the Effect of Temperature on Latrine Investment

	Change in Number of Latrines per 1,000 Households	
	(1) With State FE	(2) Without State FE
Change in number of days below 5°C	20.284* (10.809)	1.631 (8.164)
Change in number of days 5-10°C	11.746*** (3.409)	17.502*** (3.033)
Change in number of days 10-15°C	4.344** (1.741)	5.412*** (1.636)
Change in number of days 20-25°C	9.942*** (2.106)	10.075*** (1.937)
Change in number of days 25-30°C	6.801*** (2.243)	6.692*** (2.075)
Change in number of days 30-35°C	5.805** (2.569)	7.622*** (2.246)
Change in number of days above 35°C	9.231** (4.384)	8.524** (3.708)
Observations	614	614
R ²	0.547	0.116
Mean of Dep. Variable	314.341	314.341

Notes: This table reports the estimated effects of changes in the number of days in each temperature bin on changes in the number of latrines per 1,000 households, where changes are computed for each district as the difference between the mean value in the late period (2016–2019) and the mean value in the early period (2012–2015). The 15–20°C bin serves as a reference bin and is dropped from the regression. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Column 1 reports estimated effects from a specification with state fixed effects, while Column 2 reports estimated effects from a specification without state fixed effects. All columns control for changes in the number of days in each rainfall decile bin between the early and late periods. The means of dependent variables are the average differences in the number of latrines per 1,000 households between the early and late periods.

Table D5: Effect of Temperature Deviations from the Historical Mean on Latrine Investment
(Number of Latrines per 1,000 Households)

	Number of Latrines per 1,000 Households
	(1)
Number of days with temperatures >1.5 SD below the historical mean	12.055*** (2.544)
Number of days with temperatures 1-1.5 SD below the historical mean	5.107*** (1.886)
Number of days with temperatures 0.25-1 SD below the historical mean	1.156 (1.191)
Number of days with temperatures 0.25-1 SD above the historical mean	-2.128* (1.196)
Number of days with temperatures 1-1.5 SD above the historical mean	1.744 (1.531)
Number of days with temperatures >1.5 SD above the historical mean	3.485* (2.110)
Observations	3,070
R ²	0.935
Number of Districts	614

Notes: This table reports the estimated cumulative effects of temperature deviations from the historical mean, defined over 2002–2011, on latrine investment, when including up to three years of lagged temperature deviations. The sample is limited to 2015–2019 to avoid overlap between lagged temperature-deviation variables and the pre-study period used to define the historical temperature distribution. The $[-0.25, 0.25]$ standard deviation bin serves as a reference bin and is dropped from the regression. Standard errors, clustered at the district level, are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The specification includes district fixed effects and state-by-year fixed effects, and controls for the number of days in each rainfall decile bin, with up to three years of lags.

Table D6: Effect of Temperature on Latrine Use (Conditional on Ownership)

	Latrine Use Rate Conditional on Ownership (0-1)					
	(1) 1 Week	(2) 2 Weeks	(3) 1 Month	(4) 3 Months	(5) 6 Months	(6) 12 Months
Number of days below 5°C	-	-	-	-	-	-
	-	-	-	-	-	-
Number of days 5-10°C	-	-	-	-	-	0.003
	-	-	-	-	-	(0.029)
Number of days 10-15°C	-0.008 (0.011)	-0.008 (0.007)	0.002 (0.005)	0.002 (0.004)	0.005 (0.004)	0.003 (0.004)
Number of days 20-25°C	0.002 (0.009)	0.001 (0.005)	-0.000 (0.003)	0.000 (0.003)	0.000 (0.002)	-0.002 (0.003)
Number of days 25-30°C	0.005 (0.013)	-0.002 (0.007)	0.004 (0.004)	0.002 (0.003)	0.002 (0.003)	-0.001 (0.005)
Number of days 30-35°C	0.027 (0.034)	0.024 (0.022)	0.010 (0.023)	-0.004 (0.004)	0.000 (0.003)	-0.002 (0.006)
Number of days above 35°C	0.333*** (0.111)	0.488*** (0.167)	0.146** (0.059)	-0.012 (0.016)	-0.006* (0.003)	-0.013 (0.009)
Observations	874	874	874	874	874	874
R ²	0.259	0.261	0.264	0.268	0.264	0.265
Number of Households	437	437	437	437	437	437
Number of Villages	107	107	107	107	107	107
Mean of Dep. Variable	0.786	0.786	0.786	0.786	0.786	0.786

Notes: This table reports the estimated effects of temperature on latrine use rates for different reference periods. The sample is limited to households that own latrines in both survey rounds. The 15–20°C bin serves as a reference bin and is dropped from the regression. Standard errors, clustered at the village level, are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Each column includes village fixed effects, survey-round fixed effects, survey-month fixed effects, and controls for the number of days in each rainfall decile bin over the same reference period as the temperature variables. The means of dependent variables are calculated using the first survey round in 2013–2014. Dashes indicate temperature bins that cannot be estimated because no days fall into the bin for the corresponding reference period.